Beyond Yields: How Climate Change Affects Farmers Through Market Structure*

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Abstract

Climate change hurts farmers by lowering crop yields. In this paper, I show that it also hurts farmers by reshaping the markets they sell in. To document this, I exploit extreme drought shocks in the water-intensive Indian sugar industry that produces a fifth of all sugar worldwide, and where 8M smallholders sell perishable cane to oligopsonistic sugar mill intermediaries. I assemble a newly digitized 35-year administrative panel on the universe of Indian sugar mills, and a georeferenced farmer-mill transaction panel on 3M cane farmers. Using a staggered event study design, I show that droughts concentrate markets by removing the least productive mills, and trigger long-term productivity growth within survivors through managerial upgrading. Yet farm-gate cane prices fall as these gains are captured by survivors through wider markdowns. Reduced cane prices account for one half of farmers' income losses – on par with the yield losses themselves – pointing to a large role for market structure in shaping farmers' climate burden. To quantify aggregate welfare impacts, I build and estimate an equilibrium model of the Indian sugar sector, and use it to evaluate price-based policies that protect farmers.

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1 Introduction

A large and compelling literature documents that climate change imposes significant costs on agriculture.[1] Even after accounting for producer adaptation, global average yields are projected to decline by as much as 20-40% by the end of the century (Hultgren et al., 2025), implying large losses for farmers. These impacts are expected to be felt most acutely in low-income countries (Dell, Jones and Olken, 2012; Burke, Hsiang and Miguel, 2015) where a large share of the population is still reliant on agriculture. Crop yields, however, are only one part of the story when it comes to farmer welfare. What farmers earn also depends on the prices they receive for their harvest, which are typically set in markets comprising a small number of intermediaries such as traders, agribusinesses or processors.[2] A growing literature shows that these intermediaries wield substantial buyer power over farmers by significantly marking down the prices they pay below what a competitive market would deliver (Chatterjee, 2023; Rubens, 2023; Zavala, 2022; Dominguez-Iino, 2026). Prices that farmers receive are, therefore, intrinsically tied to their local market structure.

Crucially, this market structure is not fixed because intermediaries are themselves exposed to climate change, and their response impacts the prices that farmers receive. For example, a severe climate shock can push the weakest intermediaries to exit, handing greater buyer power to the ones that survive.[3] Similarly, a rise or fall in the productivity of intermediaries or their market power downstream can shift demand and move prices in either direction. Changes in local market structure can therefore either cushion farmers' losses or compound them. Yet we know little about how agricultural markets adjust in response to climate events, and how these adjustments impact farmer welfare. Filling this gap has been especially difficult because changes in long-term market structure and prices are rarely observed. This motivates the central question of the paper: How economically significant is climate-induced restructuring of agricultural markets for farmer welfare?

I take this question to the Indian sugar industry, which is the world's second largest, and produces a fifth of all sugar worldwide. In this setting, farmers sell perishable cane

[1] Some of the earlier seminal works include Mendelsohn, Nordhaus and Shaw (1994); Schlenker and Roberts (2009); Lobell, Schlenker and Costa-Roberts (2011)

[2] Traders have been documented by Bergquist and Dinerstein (2020) and Chatterjee (2023) in Kenya and India; Dhingra and Tenreyro (2021), Zavala (2022) and Dominguez-Iino (2026) study agribusinesses in six countries across Africa and South America; Avignon and Guigue (2025) and Rubens (2023) look at processors in France and China.

[3] Kochhar, Song and Sun (2024) find that higher oligopsony power in agricultural markets also blunts farmers' own capacity to adapt due to lower input usage in anticipation of lower farm-gate prices.

to local sugar mill intermediaries that produce sugar and supply downstream consumer markets. Contributing roughly 2% to India's overall GDP, the sugar sector occupies an important position in India's agricultural and industrial landscape, with 8 million smallholder farmers growing cane and 0.5 million workers employed in its 550 sugar mills. However, climate change has emerged as a real threat, in particular, due to cane's extreme water-intensity, which makes the sector highly sensitive to droughts.[4]

Droughts, the focus of this paper, are a direct consequence of meteorological and hydrological conditions such as failures of the summer monsoon, elevated temperatures that raise evapotranspiration, and the depletion of soil moisture, all of which are likely exacerbated in frequency and severity by climate change.[5] The timing of these events is arguably exogenous to local industry characteristics. Many of the most severe episodes are driven by El Niño events, which are also projected to intensify in a warming world.[6] I focus on extreme, 1-in-50-year drought events because they are severe enough to alter market structure through *direct effects* on farmers, via lower cane yields, and *indirect effects* on sugar mills, via disruptions to local supply chains.

Two features make the Indian sugar setting an ideal laboratory for this paper. First, perishability of cane ties farmers to nearby mills and creates localized markets, with a farmer typically in reach of only 5 sugar mills. This spatial concentration gives mills oligopsony power and allows me to partition cane-growing regions into distinct markets. Second, different markets experience droughts in different years, which gives quasi-experimental variation in where and when the shock lands, and in how it reshapes local market structure. Because these features characterize agricultural supply chains across a wide range of commodities and geographies[7], the paper's findings promise to be relevant for settings beyond Indian sugar.

In this paper, I make four contributions. First, I newly digitize a novel 35-year administrative panel on the universe of all sugar mills in India and supplement it with a georeferenced farmer-mill transaction panel on 3 million cane farmers and a survey of 220 sugar mills. This data allows me to document the structure and evolution of cane markets throughout India on an unprecedented scale. Second, exploiting spatial and temporal variation in extreme drought events, I provide empirical evidence that exposure to droughts causes the least productive sugar mills to exit, and their farmers to reallocate to

[4]Mention press release on vulnerability of Indian sugar sector to droughts

[5]Cite papers on causes and increasing frequency of droughts in India

[6]Cite papers on (1) El Niño - India droughts link; (2) El Niño impacts more severe under climate change

[7]Mention other commodities and regions - like coffee in Rwanda, Costa Rica (Macchiavello et al.); maize in Kenya (in Bergquist and Dinerstein); tobacco in China (Rubens); cash crops in Peru (Zavala)

nearby survivors. These surviving mills reorganize their labour and improve cane quality management, allowing them to observe long-term productivity gains. Strikingly, within-mill cane prices see a decline indicating that internal productivity gains are captured by surviving mills through amplified markdowns. Third, I develop an equilibrium framework of the Indian sugar sector to rationalize the empirical findings. I combine a nested discrete choice system for farmers, who choose among crops and among sugar mills if they grow cane, with oligopsonistic pricing among mills. Fourth, I estimate this model using mill-level and farmer-level data to quantify how drought-induced selection and within-mill productivity growth contribute to aggregate productivity gains among sugar mill intermediaries. More importantly, I quantify farmer welfare, and evaluate how the market responds to public intervention in the form of price-based policies.

To examine how extreme droughts reshape local market structure and impact cane farmers, I assemble several novel datasets that allow me to link sugar mill operations and farm-gate prices with drought exposure. First, I digitize archival records from the Indian Directorate of Sugar, which document comprehensive information on procurement, production, and managerial and technological practices for the universe of Indian sugar mills from 1990 to 2025. Second, through a close partnership with the Sugar Commissionerate in Maharashtra – India's largest sugar producing state – I obtain first-time access to a georeferenced panel on farmer-mill transactions covering all 3 million cane farmers in the state from 2020 to 2025. Third, I conduct a survey of 220 sugar mills in Maharashtra and elicit information on drought risk, farmer-mill relationships and climate adaptation investments. Next, I link these administrative records to each mill's financial data from 1993 to 2023 using common identifiers with the Annual Survey of Industries (ASI). This allows me to identify, for the first time, every plant in the ASI for a large and important industry.[8] Finally, I construct Standardized Precipitation-Evapotranspiration Indices (SPEI) – my primary measure of extreme droughts – for each market-year, using high resolution data from the Drought Atlas of India (Chuphal et al., 2024).

Using this combined data, I document three stylized facts that directly motivate my empirical strategy and findings. First, trade in cane is highly localized, giving rise to spatially segmented markets. Nearly 80% of all trade occurs within 25 km of a mill location and virtually all trade exhausts within 50 km. This is because cane's sugar content deteriorates rapidly after harvest, leaving mills with only a 12–24 hour window to process it. Second, mills have price-setting power over farmers in these local cane markets but behave as price-takers in the sugar market. This is driven by policy wherein the central government

[8]The ASI does not disclose plant names or any identifier other than an anonymous plant ID.

tightly controls mill-level sugar releases to stabilize consumer prices. Third, mills differ enormously in observed productivity, measured by sugar recovery. The most productive mills extract more than three times as much sugar from a given quantity of cane as the least productive, who also earn thinner margins per tonne of cane crushed. This makes the least productive mills most vulnerable to extreme droughts, setting the stage for my empirical strategy and results.

My empirical strategy exploits spatial and temporal variation in 1-in-50-year drought events which correspond to SPEI values lower than -2. To quantify market- and mill-level adjustments to droughts, I implement an event-study design using the Sun and Abraham (2021) estimator to address concerns about heterogeneous treatment effects in staggered difference-in-differences settings. The empirical results reveal four key findings. First, extreme droughts raise local market concentration, driven by the exit of least productive sugar mills and the reallocation of their farmers to nearby survivors. I measure market concentration using a Herfindahl-Hirschman Index (HHI) in cane procurement shares which remains elevated even a decade after the shock. Second, surviving mills reorganize their labour and improve cane quality management by harvesting cane at better maturity. This allows them to raise sucrose content in procured cane by 4 percent and observe an increase in sugar recovery – the observed measure of productivity – by 5 percent over the subsequent decade. The differential impact is driven by cultivation of newly released, drought-resistant cane varieties that are, in part, distributed by surviving mills to their catchment farmers. Improvement in within-mill management practices, therefore, explain more than 80 percent of the increase in long-term mill productivity. Third, and strikingly, within-mill cane prices see a 2-3% decline in the long-term relative to mills in control markets, suggesting that productivity gains are captured by surviving mills through amplified oligopsony power in cane markets. A decomposition of aggregate impacts on farmer cane income into a *yield* component and a *price* component reveals that reduced cane prices explain one half of farmers' income losses – on par with the yield losses themselves. Together, these results point to a large role for market structure, a previously unexplored channel, in shaping farmers' long-term climate burden.

To interpret these findings, I develop and estimate an equilibrium model of the Indian sugar sector that links oligopsonistic sugar mills' demand for cane with upstream farmer supply. On the demand side, sugar mills operate a Leontief technology between cane and a labour–capital aggregate. This is to account for the fact that cane cannot be substituted with either labour or capital in order to produce sugar. A tonne of cane yields sugar at an exogenous sugar recovery rate, while labour and capital produce processing capacity

subject to Hicks-neutral productivity. Mills choose labour to maximize variable profits, and through their cane demand, set farm-gate cane prices. These are marked down from the marginal revenue product of cane subject to a statutory price floor.

On the supply side, I model farmer decisions using a nested discrete-choice framework that captures two margins of adjustment. In the outer nest, a farmer chooses which specific agricultural commodity to grow. In the inner nest, a farmer growing cane chooses which mill to sell to, with choice shaped by the offered price, distance to the mill, and observed mill characteristics. This nested structure generates an own-price supply elasticity with two margins – cross-mill substitution (stealing farmers from rival mills) and cross-crop substitution (pulling farmers into cane from alternative crops). This elasticity carries the paper's market-power mechanism: when droughts increase market concentration through mill exit and farmer reallocation, a surviving mill's within-cane share rises, its residual supply becomes less elastic, and its markdown widens.

Estimation proceeds in two stages. First, I estimate the production function using a control-function approach with endogenous exit (Olley and Pakes, 1996; Ackerberg, Caves and Frazer, 2015). To address the Doraszelski and Jaumandreu (2021) critique that input demand may shift due to either productivity or markdowns, I use the nested farmer supply model to control for markdown variation with observed prices of sugar and cane, and mill-level market shares, thereby restoring the scalar-unobservable assumption. Next, I estimate the supply elasticities using crop-level and mill-level shares with instruments for prices. As expected, I find that mills are closer substitutes for one another than cane is for competing crops, suggesting substantial crop switching costs. I also find the cross-crop elasticity estimate to be in line with values from the agricultural trade literature.[9]

The estimates reveal that sugar mills hold considerable buyer power over farmers: the average mill paid its farmers 80% of the marginal revenue product of cane. This markdown level is comparable with figures obtained for other agricultural goods in the literature.[10] Using the estimated TFP and cane price markdowns, I examine how drought-induced market restructuring affected both mill productivity and buyer power. I find that droughts increased within-mill TFP by X percent on average, compared to a 5 percent rise in sugar recovery (equivalently, the marginal product of cane) documented earlier. Droughts also led to increased exertion of oligopsony power in cane markets: markdowns

[9] My estimated value of 2.12 sits in between the range of 1.5-2.5 documented in Sotelo (2020); Bergquist and Dinerstein (2020); Costinot, Donaldson and Smith (2016); Dominguez-Iino (2026).

[10] Average inverse markdowns in Avignon and Guigue (2025); Dominguez-Iino (2026); Zavala (2022); Crespi and Sexton (2005) range between 0.66-0.9. My inverse markdown figure of 0.8 falls in the middle of this range.

increased on average by Y percent, a decade after the shock.

Next, I use the model to quantify the impact of extreme droughts on farmer welfare. First, I decompose the productivity gains from drought-induced restructuring into a *between-firm* resource reallocation component and a *within-firm* improvement component. Lower cane supply reduces variable profits of mills, pushing the weakest ones into losses. Surviving mills, who inherit farmers from exiting mills, respond by improving management practices and observe a rise in productivity. Mills whose productivity is too low to survive, even with these improvements, exit the market. Both responses – productivity improvements among surviving mills and the exit of unproductive ones – raise aggregate productivity by 18 percent. A decomposition reveals that approximately 25 percent of this increase comes from the exit of low-productivity sugar mills, with the remaining 75 percent stemming from within-firm productivity improvements. This highlight the role of managerial improvements in contributing to productivity gains after droughts. I pair this decomposition with a full welfare accounting for both sides of the market, tracing how each channel maps into mill profits and farmer surplus. Results under estimation at present. Second, I evaluate how the market responds to public intervention in the form of price-based policies. Specifically, I focus on the role of statutory price floors in protecting farmers from increased oligopsony power.

Related Literature: This paper adds to four strands of the literature. First, it speaks to a large body of work documenting the impacts of climate change on agriculture, which has focused overwhelmingly on the physical dimension of production, such as yield and output (Schlenker and Roberts, 2009; Lobell, Schlenker and Costa-Roberts, 2011; Hultgren et al., 2025), or land values (Mendelsohn, Nordhaus and Shaw, 1994; Deschênes and Greenstone, 2007). These damages are expected to fall hardest on poor, hot countries, where agriculture is a larger share of economic activity and adaptive capacity is more limited (Dell, Jones and Olken, 2012; Burke, Hsiang and Miguel, 2015). I depart from this literature by studying climate change's effect on farmers through a channel that is largely unexplored – the price they receive for their produce. Because most farmers do not sell into thick, competitive markets but to a small number of local intermediaries who wield substantial buyer power (Bergquist and Dinerstein, 2020; Chatterjee, 2023; Zavala, 2022; Rubens, 2023; Dominguez-Iino, 2026; Dhingra and Tenreyro, 2021), the incidence of a climate shock depends on how it transmits through these intermediaries — and how the shock reshapes them. I make this concrete by studying one of the developing world's most economically important agricultural supply chains, the Indian sugar industry, where millions of farmers sell water-intensive cane to a concentrated set of processing

mills, and show that the price channel can leave farmers no better off even as the intermediaries adapt and see productivity growth.

Second, this paper adds to a growing literature examining how natural disasters and climate events affect firm outcomes. Earthquakes have been shown to raise short-run productivity through creative destruction but with effects that fade over time (Cole et al., 2019; Tanaka, 2015). Floods affect capital accumulation and employment growth within firms, with heterogeneous effects depending on asset structure (Leiter, Oberhofer and Raschky, 2009). Extreme temperatures reduce manufacturing productivity through their effects on labour and factor reallocation (Zhang et al., 2018; Somanathan et al., 2021). Hurricanes cause lasting output losses and reshape local economic geography (Hsiang and Jina, 2014; Deryugina, Kawano and Levitt, 2018; Basker and Miranda, 2018). And more broadly, well-managed firms have been shown to be more resilient to environmental shocks (Keiller and Van Reenen, 2024). I add to this literature by documenting the effects of extreme droughts, an understudied disaster type in the firm-level literature, on firm performance, upgrading, and market structure in a developing-country agricultural processing industry. While most of the existing literature focuses on the direct damage to affected firms, my results highlight an indirect channel: drought-induced restructuring triggers endogenous productivity gains among *surviving* firms through a market structure mechanism that operates over a decade, well beyond the duration of the drought itself.

Third, an emerging literature documents the secular rise of market power across developed economies and investigates its drivers. De Loecker, Eeckhout and Unger (2020) document that aggregate markups in the US rose from 21 percent above marginal cost in 1980 to 61 percent by 2016. Autor et al. (2020) attribute rising concentration to "superstar firms" that leverage technological advantages into dominant market positions. Grullon, Larkin and Michaely (2019) document rising profitability alongside rising concentration across US industries. The drivers explored in this literature are primarily technological change, globalisation, and regulatory barriers. I propose a new mechanism: *climate change*. As extreme weather events become more frequent and more intense, they will increasingly cause selective firm exit, particularly in climate-sensitive industries with high sunk costs and spatially segmented markets, generating persistent concentration increases that are not easily reversed by entry. My results show that even a single extreme drought can persistently reshape local market structure, with concentration and input price markdowns remaining elevated a decade later. If drought frequency doubles under climate projections, the steady-state level of concentration and markdowns will be permanently higher.

This positions climate change as a structural driver of market power in agricultural processing and potentially in other climate-exposed industries with similar characteristics.

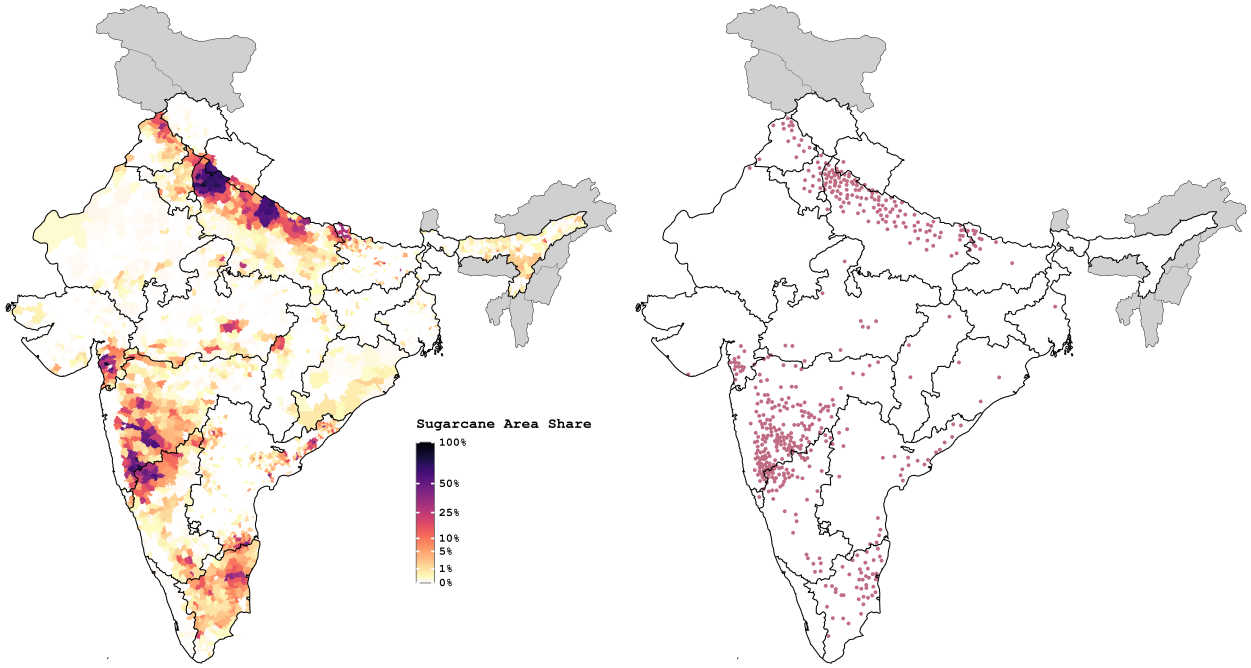
Finally, a large body of work has documented how external shocks raise aggregate productivity through *between-firm* reallocation: the exit of low-productivity firms and the expansion of efficient survivors (Melitz, 2003; Foster, Haltiwanger and Krizan, 2001; Foster, Haltiwanger and Syverson, 2008; Syverson, 2011; Collard-Wexler and De Loecker, 2015; Bartelsman, Haltiwanger and Scarpetta, 2013). A separate literature has shown that external shocks also trigger *within-firm* productivity gains through organizational change, technology adoption, and managerial upgrading (Bustos, 2011; Lileeva and Trefler, 2010; Bloom, Draca and Van Reenen, 2016; Verhoogen, 2008; Kotia, 2025). Substantial progress has been made in understanding each channel individually. I contribute by establishing a causal link between them: I show that the *between-firm* reallocation process endogenously triggers the *within-firm* upgrading response through a volume windfall mechanism. Unlike prior work where both channels operate as parallel responses to the same external shock, in my setting the first channel is a necessary precondition for the second - without exit there is no reallocation, without reallocation there is no volume windfall, and without the volume windfall the marginal return to upgrading remains low.

The remainder of the paper is organised as follows. Section 2 provides institutional background on the Indian sugar industry. Section 3 describes the data sources and data collection exercise. Section 4 presents motivating stylized facts. Section 5 outlines the empirical strategy and presents the event-study results. Section 6 develops the structural model. Section 7 discusses identification and estimation. Section 8 presents counterfactual simulations. Section 9 concludes.

2 Institutional Background

India is the world's second-largest producer of sugar behind Brazil, accounting for a fifth of global production. The industry occupies a central position in the Indian economy – it sources cane from roughly 8 million smallholder farmers, supporting 50 million livelihoods, and directly employs half a million industrial workers.[11] In the process, it contributes to approximately 2 percent of India's GDP. Close to 550 sugar mills intermediaries currently operate in 16 states across the country, although 80% of national sugar output is concentrated in the three major producing states of Maharashtra, Uttar Pradesh and Karnataka. Figure 1 shows the intensity of cane cultivation and location of sugar mills across India in 2015-16.

Figure 1: Sugar Production in India



Panel A: Cane Cultivation, 2015-16 **Panel B:** Location of Sugar Mills, 2015-16

Notes: Panel A shows block-level share of cropped area under cane cultivation in 2015-16. Panel B shows the location of operational sugar mills in India in the 2015-16 season. *Source:* Indian Agriculture Census 2015-16, and plant-level records from the Indian Directorate of Sugar.

2.1 Indian Sugar Value Chain

The Indian sugar value chain is visualized in Figure 2. At the top of the chain, 8 million smallholder farmers cultivate cane on plots averaging 0.8 hectares [12]. These farmers

[11] For details, see https://www.pib.gov.in/FactsheetDetails.aspx?Id=150971

[12] This figure comes from the Indian Agriculture Census 2015-16.

sell their cane to approximately 550 sugar mills, with each mill employing close to 1,000 workers and sourcing cane from nearly 15,000 farmers on average. In this setting, sugar mills therefore become the primary processing intermediaries that cane farmers transact with[13]. Farmers are therefore atomistic and price-takers in cane markets. Crucially, it is the mill, not the individual farmer, that designs and coordinates the harvesting schedule during the crushing season, which typically runs for 5 months from November to April. Mills determine which fields are cut on which days to manage the flow of cane to the factory and to ensure that cane is crushed near peak maturity. Cane is then crushed to extract juice, and finally processed into raw or refined sugar. The sugar is then sold to wholesalers and retailers who supply the consumer market.

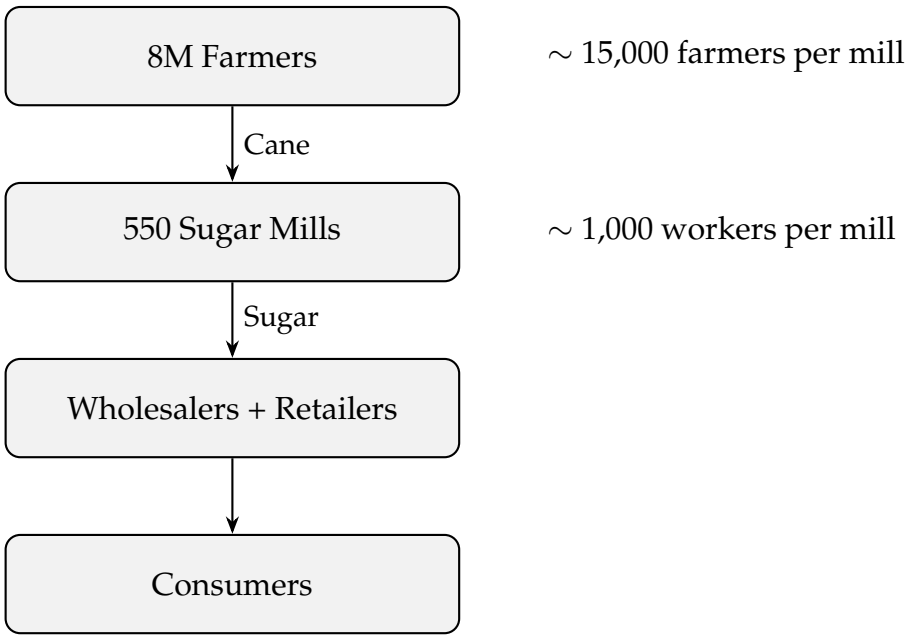


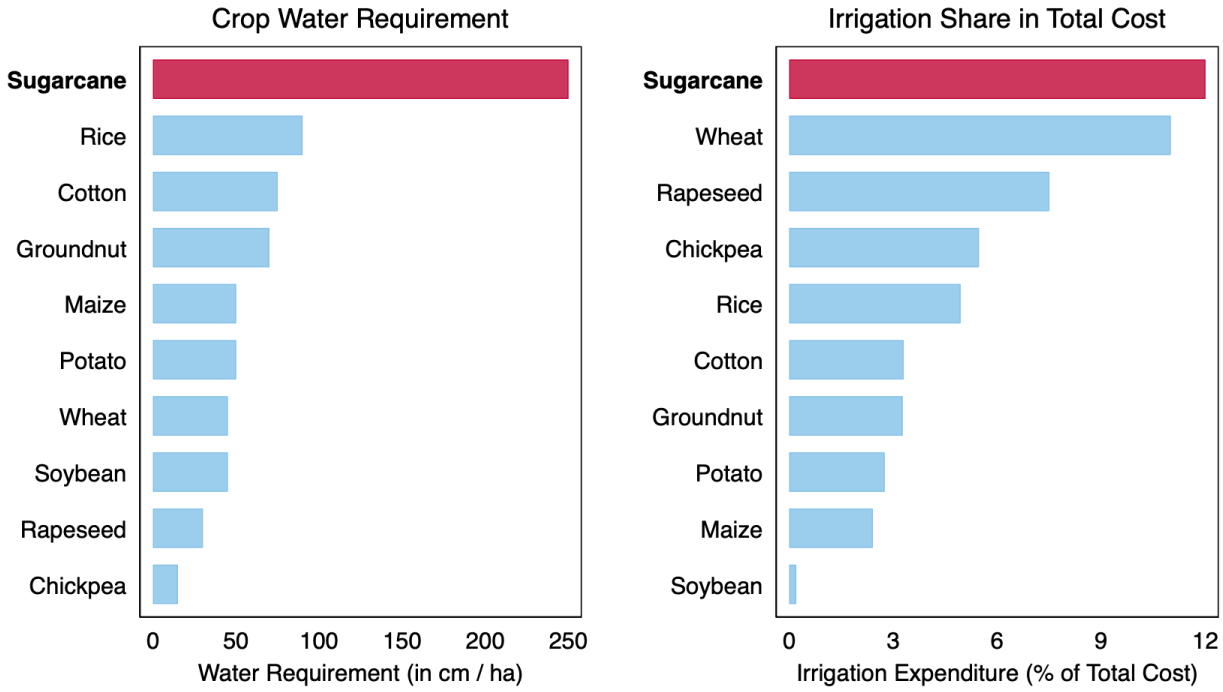
Figure 2: The Indian Sugar Value Chain

Mill ownership takes two primary forms: cooperatives and private enterprises. Cooperatives, which are particularly prevalent in Maharashtra and Karnataka, are nominally owned by their farmer-members but managed by elected boards. A natural concern is that cooperatives may face different optimization objectives than private mills, for instance, maximizing member welfare rather than profits (Hansmann, 2000). However, farmer-level data from Maharashtra in Figure A.1 reveals that less than a quarter of cane suppliers to a typical cooperative are registered member-farmers; the remaining three-

[13]Over 80% of all cane produced is utilized for sugar production, with the rest sold to small jaggery producing units or used for self-consumption. I introduce these as an outside option in the inner nest of my nested discrete choice farmer supply model in Section 6.

fourths are non-member suppliers with no ownership stake. Since cooperatives must offer competitive cane prices to attract this large non-member majority who face the same outside options as farmers supplying to private mills, the pricing behaviour of cooperatives in the cane market is likely to resemble that of profit-maximizing private mills. As such, the model that I develop in Section 6 treats ownership status as an observable characteristic entering farmer mill-choice but does not endogenize the governance structure.

2.2 Water Intensity and Sensitivity to Droughts



Panel A: Crop Water Requirement

Panel B: Irrigation Share in Total Cost

Figure 3: Cane's water intensity makes sugar mills vulnerable to droughts

Notes: Panel A shows crop water requirement over the growing season for 10 major crops, while Panel B shows their share of total cultivation cost spent on irrigation. *Source:* Indian National Bank for Agriculture and Rural Development (NABARD), and the Cost of Cultivation Surveys.

The Indian sugar industry's vulnerability to droughts stems from the extraordinary water intensity of cane cultivation. Cane requires approximately 250 centimetres of water per hectare over its 12 month growing cycle (Panel A of Figure 3), making it the most water-demanding crop in Indian agriculture. For comparison, sugarcane's water requirement is nearly thrice that of rice, the next most water-intensive major crop. Cane also stands

out in terms of expenditure incurred on irrigation, with 12% of its total cost of cultivation spent on irrigation through groundwater or canals (Panel B of Figure 3). This is despite the fact that a sizeable part of its growing cycle overlaps with the monsoon season, as opposed to wheat which is grown during the dry season. Crucially, the water intensity extends beyond cultivation into the processing stage. Sugar mills, on average, require 20 cubic metres of water to produce one metric tonne of sugar (Figure A.2). Both the upstream cultivation and downstream processing stages of the supply chain are thus acutely dependent on water, making the entire industry highly sensitive to droughts. This sensitivity is not merely theoretical – it is a recurring feature of India's economic landscape that regularly makes national headlines as can be seen in Figure A.3.

The most widely used measure of mill productivity in the industry is *sugar recovery* – the kilograms of sugar extracted per 100 kilograms of cane. This measure is determined by two inputs - sugar content in cane and the mill's extraction efficiency, as illustrated in Figure **??**. On the input quality side, two factors influence sugar content in cane: cane maturity at harvest, measured by brix percentage in cane's primary juice (the concentration of dissolved solids, of which 70-85% is sucrose), and the cane variety (improved varieties contain higher sucrose within the brix and lower impurities). On the processing side, extraction losses – sugar lost during crushing, clarification and crystallisation – reduce the recoverable share. Sugar recovery thus equals sugar content in cane minus extraction losses. Mills can therefore improve recovery through two distinct channels: improving the quality of cane arriving at the factory gate (the input quality channel), or reducing losses during extraction (the processing efficiency channel).

3 Data

I assemble four datasets that jointly capture mill-level operational outcomes, cane input prices, farmer supply behaviour, and drought shocks at resolutions fine enough to identify causal effects within local sugar markets. Three novel data contributions underpin the analysis. First, I digitise archival plant-level records from the Indian Directorate of Sugar - the closest available census of the Indian sugar manufacturing sector - covering the universe of all sugar mills from 1990 to 2025 and yielding a panel of unprecedented length and coverage. Second, I obtain first-time research access to the complete universe of farmer-mill cane transaction records maintained by the Maharashtra Sugar Commissionerate, providing georeferenced data on approximately 3 million unique cane farmers. Third, I develop a novel matching algorithm to link the Directorate's plant-level records

to the Annual Survey of Industries (ASI), enabling a full mill-level price and financial analysis for the first time in the Indian sugar industry literature. This section describes each dataset in turn.

3.1 Plant-Level Data

3.1.1 Archival Mill-Level Records

The core of the database is a mill-level panel constructed from the Yearbook and Directory of Indian Sugar Factories, published annually by the Indian Directorate of Sugar. The Yearbook constitutes the closest available census of the Indian sugar manufacturing sector, covering all registered and operational mills in every crushing season. I obtain access to and digitise these archival records spanning 1990 to 2025 through an institutional relationship developed with the Directorate. I supplement these with plant-level records from India's two largest sugar producing states - Maharashtra and Uttar Pradesh - that together account for over 70% of the country's sugar output. Records obtained from the state-level sugar departments provide an additional cross-check on coverage. The final sample consists of 886 firms in 750 plant locations across 20 states.

For each mill-year, the panel records administrative and manufacturing data for each step of the production process. Key variables of interest include total cane procured, sugar produced, crushing duration, installed capacity, capacity utilisation, cane maturity at harvest, sugar content in cane, overall sugar recovery rate, extraction efficiency, and water use. The sugar recovery rate - defined as sugar produced per unit of cane crushed, and expressed as a percentage - is the observable measure of mill-level productivity. Crucially, recovery can be decomposed into two independent components: Pol% (which measures sugar content in cane - a field-level variable reflecting variety choice and harvest timing), and Mill Extraction and Boiling House Recovery (which measure extraction efficiency - a factory-level technology variable). This decomposition allows the analysis to separately identify whether recovery gains arise from field-level managerial improvements or factory-level technological investments - the key distinction for the paper's mechanism. Operational status is coded as working, suspended, or permanently exited, with permanent exit assigned when a mill is non-working for more than three consecutive years with no subsequent resumption, or when corporate records confirm acquisition or liquidation.

3.1.2 Cane Variety Technology

The panel also records the varieties of cane procured by each mill in each season, providing a direct measure of varietal composition over time. I link these variety names to agronomic attributes – potential sucrose content, yield, maturity class, and agro-climatic zone using variety performance data from the All India Coordinated Research Project (AICRP) on sugarcane and Vasantdada Sugar Institute's variety evaluation database. Drought and stress tolerance classifications are obtained from the ICAR-Indian Sugarcane Research Institute. This enables construction of an annual mill-level cane variety index.

3.1.3 Online Mill Survey

To supplement the administrative records with information on management practices and organizational structure not captured in the panel, I conduct an online survey of 220 sugar mills in Maharashtra, circulated through the Maharashtra Sugar Commissionerate. The survey covers drought risk and adaptation, harvest scheduling practices, extension staff deployment, seed distribution programmes, and recent capital investments - the management inputs that the model identifies as the channel for within-firm productivity upgrading.

3.2 Financial Data

3.2.1 Annual Survey of Industries

Cane input prices and financial structure at the mill level are drawn from the Annual Survey of Industries (ASI), conducted annually by India's Ministry of Statistics and Programme Implementation. The ASI covers all factories registered under the Factories Act, 1948, and provides data on revenues, costs, wages, capital, material inputs, and output - making it the principal source of industrial statistics in India and the only source providing a standardized, all-India financial panel for manufacturing plants.

A central challenge is that the ASI does not identify factories by name; it uses an internal factory code not publicly linked to plant identities, and these codes change with ownership transfers or re-registration. I overcome this challenge through a matching algorithm that exploits variables common to both the archival plant records and ASI. This yields a matched panel for the near-universe of sugar mills from 1993-2023, enabling a full cane price and financial analysis at the individual mill level for the first time in the Indian sugar industry literature.

3.2.2 Maharashtra Financial Data

For Maharashtra, three additional sources provide financial data. First, the Sugar Commissionerate publishes annual mill-level price reports, recording the actual cane price paid to farmers by season. Second, the Vasantdada Sugar Institute maintains detailed cost-of-production records for sugar mills in Maharashtra from 1998–2023, serving as an independent validation layer for ASI-derived cost variables and as a primary source for mills outside the ASI match. Third, I draw on mill licence application records maintained by the Commissionerate covering 1990-2025. These records document the year of application, the year of first crushing, and for mills that entered, directly observed entry costs decomposed into land acquisition, machinery, and building costs. This is, to my knowledge, the only source providing directly observed sunk entry costs for sugar mills in India, enabling identification of the entry cost parameter from data rather than from equilibrium restrictions alone.

3.3 Farmer-Mill Transactions Data

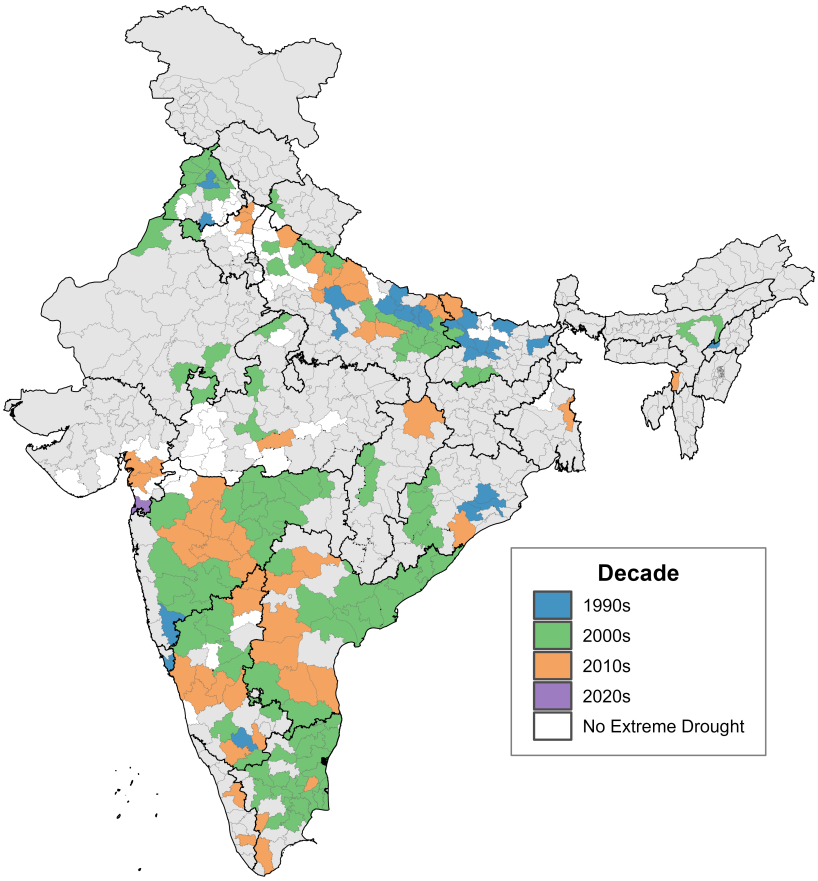
The Maharashtra Sugar Commissionerate maintains, under the state's mandatory cane registration system, a complete record of every farmer-mill cane transaction in Maharashtra. For each crushing season from 2020 to 2025, the records document, for every farmer supplying cane to every mill: the farmer's village, cane variety grown, area under cane, maturity class, and cooperative membership status. I georeference these records by matching Commissionerate village names to census village identifiers through a multi-stage fuzzy matching pipeline. The final output covers approximately 3 million unique farmer observations across 220+ mills and 16,000 villages.

3.4 Weather Data

The treatment variable throughout the main analysis is an indicator for an extreme drought at the district level, constructed from the Drought Atlas of India (Chuphal et al., 2024) - a high-resolution dataset providing monthly Standardised Precipitation Evapotranspiration Index (SPEI) values across India from 1901 to 2021. The SPEI accounts for atmospheric water demand, temperature-driven evapotranspiration, alongside water supply, making it better suited to capturing crop-water stress than simpler precipitation measures such as SPI. The SPEI values are constructed using the Hargreaves method, which requires only minimum and maximum temperature and extraterrestrial radiation as inputs, and is considered highly reliable for Indian climatic conditions.

I use a 12-month accumulated SPEI measured from January to December, chosen to match the sugarcane cultivation cycle: cane is typically planted in January and harvested in December, so a January-to-December window captures the full annual water balance relevant to the crop. An extreme drought is defined as a district-year in which SPEI falls at or below -2, corresponding approximately to a 1-in-50 year event.

Figure 4: Drought Incidence across Sugar Producing Districts



Construction proceeds in three steps. First, I aggregate the Drought Atlas's native-resolution data to the block level within each district. Second, if any block within a district records SPEI ≤ -2, that district is classified as drought-affected in that year. This is justified by the fact that mills source cane from catchment areas of approximately 50 kilometres, so a block-level extreme drought will affect all mills in the district. Third, to avoid complications from multiple drought incidences within the same district over time, I retain only the single most severe drought for each district. Severity is defined as the year in which the largest share of the district's area falls below SPEI ≤ -2 - that is, the drought with the broadest spatial coverage. This ensures each treated district has one cleanly defined

treatment year, simplifying the event study design.

Over the sample period, the data records 174 extreme drought events affecting 226 distinct districts across 20 cane-producing states, with staggered timing across 25 distinct years that provides the identifying variation. Figure 4 maps this spatial and temporal incidence of extreme droughts across Indian districts, colour-coded by the decade in which each district experienced its most severe event. The map reveals rich variation across both space and time: drought-affected districts are distributed across all major sugar-producing regions, with no single state or region bearing the shock exclusively.

3.5 Supplementary Data

I complement the core datasets with several agricultural, irrigation, and geographic sources used for controls, the farmer supply model, and robustness checks. The quinquennial Agriculture Census provides block- and district-level crop area and irrigated area. The Minor Irrigation Census rounds provides data on wells, tube well depths, and irrigated command area, used as controls for groundwater dependence. Farmer-level Cost of Cultivation Surveys (2000-2020) provide input costs, revenue and yields for major crops including sugarcane. The ICRISAT District-Level Database (to 2017), supplemented by Area, Production and Yield (APY) reports from the Directorate of Economics and Statistics for subsequent years, provides district-level crop yields, output, acreage, and prices for all major crops entering the Fréchet outer nest. FAO-GAEZ potential crop yields under different agro-climatic conditions are used to construct land suitability parameters in the supply model. Finally, Central Ground Water Board's (CGWB) observation-well-level groundwater depth measurements are used in the event study and as a control for irrigation dependence.

4 Stylized Facts

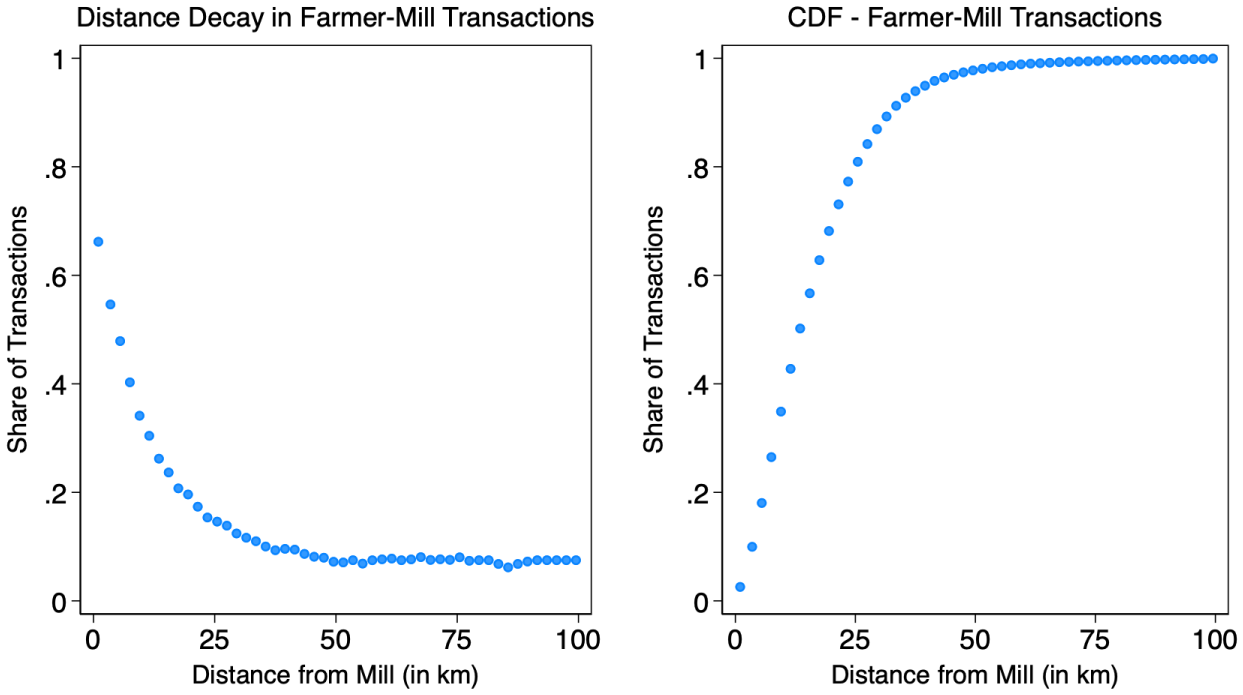
Before turning to the event study results, I present three stylized facts that establish the institutional conditions under which the paper's mechanism operates.

Fact 1: Trade in cane is highly localized, generating spatially segmented markets.

Cane markets are spatially segmented to a degree that is unusual even among agricultural commodities. Using georeferenced farmer-mill transaction data, I document extreme distance decay in farmer-mill relationships. Figure 5, panel A plots the share of farmers

selling to a given mill as a function of the distance between them. Among farmers located within 5 kilometres of their closest mill, 70 percent sell their cane to that mill, and this share drops to less than 10 percent by 50 kilometres. Figure 5, panel B presents the cumulative distribution function: roughly 80 percent of all farmer-mill transactions take place within 25 kilometres of a mill and virtually all trade in cane exhausts within 50 kilometres of a mill.

Figure 5: Trade in cane exhausts within 50 kms of a mill



Panel A: Distance Decay

Panel B: CDF - Distance Decay

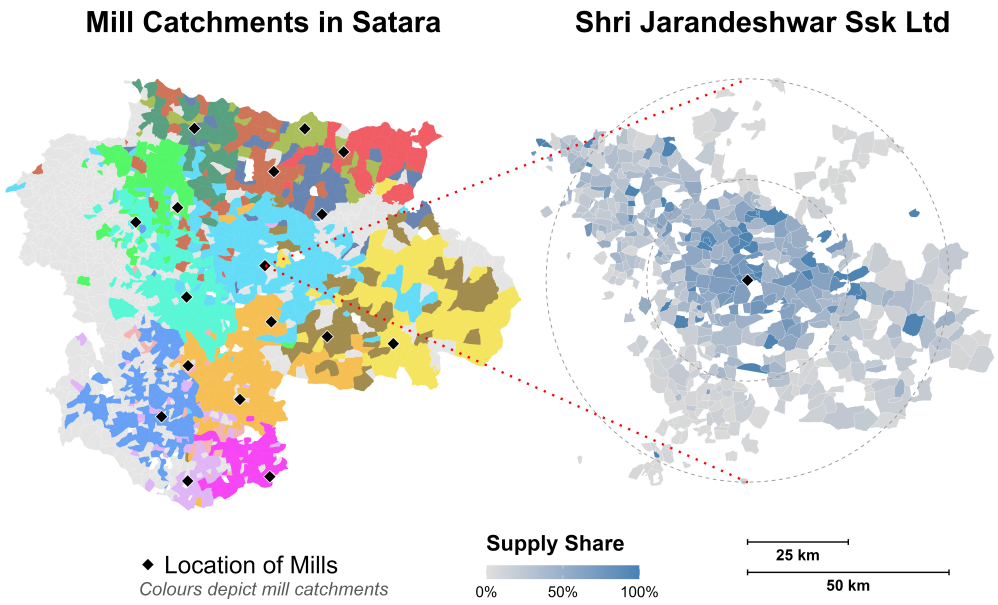
Notes: Panel A shows the share of farmers selling to a given mill as a function of the distance between them. Panel B plots the CDF. *Source:* Farmer-Mill Transactions from the Maharashtra Sugar Commissionerate.

This extreme spatial concentration arises from the rapid post-harvest deterioration of sugar content in cane. Once cut, cane loses recoverable sugars rapidly due to microbial activity and chemical inversion of sucrose into non-recoverable reducing sugars – glucose and fructose. This perishability creates strong incentives for mills to source cane from nearby farms and crush cane within 12-18 hours of harvesting. Such a highly localized nature of trade in cane gives rise to spatially segmented markets as illustrated in Figure 6. Panel A maps the district of Satara in Maharashtra, with each village colour-coded by the mill that receives the largest share of its cane and reveals sharp spatial partitioning.

Panel B zooms into the catchment area of a single mill, showing that the mill's procurement share across surrounding villages decays steeply with distance – consistent with the distance gradient documented in Figure 5.

This fact has three implications for the paper. First, it defines the relevant market: cane markets are local, and market concentration measures defined at the district level capture meaningful variation in competitive conditions. I therefore adopt the district as the unit of a local market throughout the analysis. This choice is validated by farmer-mill transaction data from Maharashtra, which shows that cross-district trade in cane accounts for only 10 percent of all transactions (Figure A.4) – the vast majority of cane procurement occurs within district boundaries. Second, it ensures that when a mill exits, its farmers are reallocated to nearby surviving mills. Third, it creates oligopsony power: with only a fixed set of mills within farmers' economical transport distance, each mill faces a local supply curve with limited elasticity, giving it price-setting power over cane procurement.

Figure 6: Cane markets are spatially segmented



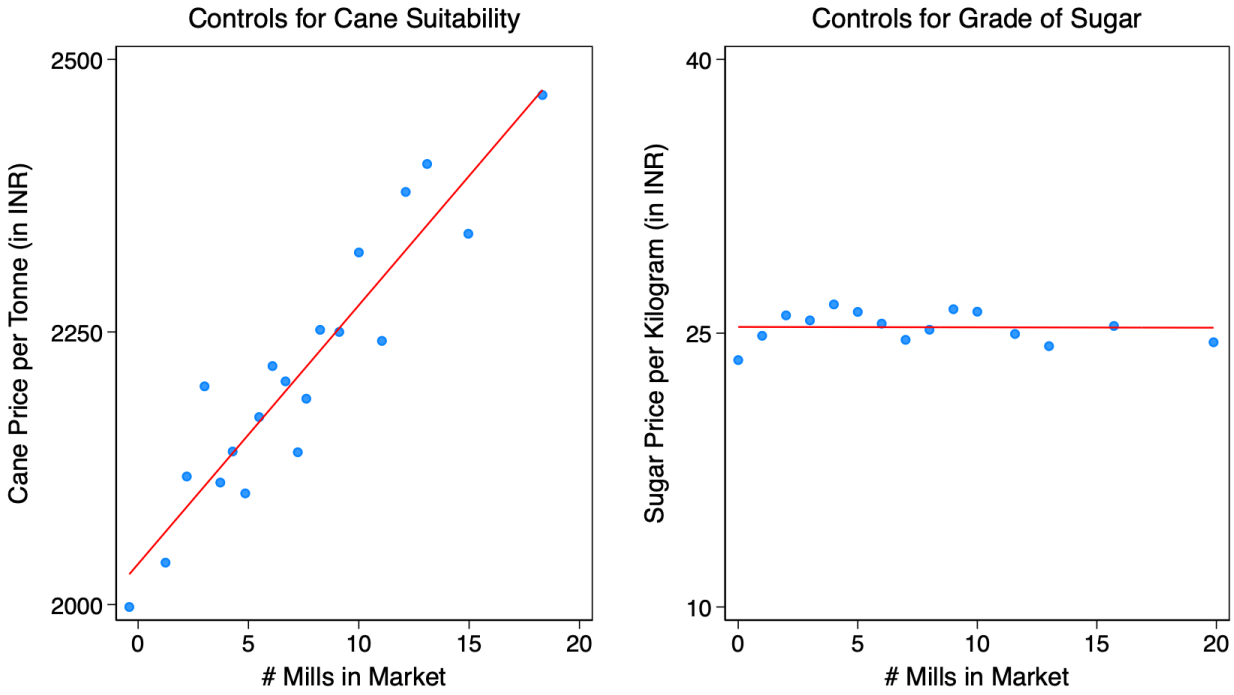
Notes: The plot on the left shows the distribution of mill catchments in the district of Satara, Maharashtra. The plot on the right zooms into the catchment for one mill, showing the mill's procurement share across supplying villages in a 50 km radius. *Source:* Farmer-Mill Transactions from the Maharashtra Sugar Commissionerate.

Fact 2: Mills set cane prices, but are price-takers in sugar markets.

Cane markets are spatially segmented because cane is bulky, perishable, and costly to transport. This feature creates local oligopsonies: a given farmer's choice set is limited

to the few mills within economical transport distance. To protect farmers, cane prices are subject to regulatory price floors in the form of a centrally determined Fair and Remunerative Price (FRP) and, in several states, the higher State Advised Price (SAP).[14] However, these floors are not binding in practice: more than 90% of all observations in the data see farm-gate cane prices above the price floor. Panel A of Figure 7 demonstrates the resulting oligopsony power: cane prices paid to farmers are higher in markets with a greater number of mills, after controlling for cane quality and suitability. Moving from near-monopsony to a competitive market of 15 mills raises cane prices by approximately 25 percent, suggesting price-setting power by mills in cane markets.

Figure 7: Mills have price-setting power in cane markets



Panel A: Cane Prices

Panel B: Sugar Prices

Notes: Panel A shows a binscatter plot for farm-gate cane prices against local market structure after adjusting for cane quality and suitability. Panel B shows a binscatter plot for ex-mill sugar prices against local market structure after adjusting for the grade of sugar produced. Data is for year 2010. *Source:* Plant-level Records from the Indian Directorate of Sugar and the Annual Survey of Industries.

On the output side however, the picture is starkly different. Sugar prices are effectively

[14]Four states - Uttar Pradesh, Punjab, Haryana and Uttarakhand set their own State Advised Price (SAP), which specifies two to three distinct prices according to the cane variety a farmer grows; the remaining states follow the centrally determined Fair and Remunerative Price (FRP), which combines a baseline price with a premium that increases in the mill's sugar recovery.

determined by government regulation through a system of monthly release quotas that control how much sugar each mill can sell in a given month. An example of such a monthly sugar release allocation can be seen in Figure A.5 for sugar mills in Punjab. Since each mill's saleable quantity is administratively fixed, no individual mill can influence the market price by expanding or withholding supply. Panel B of Figure 7 confirms this: ex-mill sugar prices in the year 2010 show no systematic relationship with local market structure after controlling for the grade of sugar produced. Sugar prices are seen hovering around Rs. 25-26 per kilogram regardless of how competitive a mill's local market is, as measured by the number of mills located in the district. Market power, absent on the output side but pronounced on the input side, is a defining feature of the industry and motivates the oligopsony model I develop in Section 6.

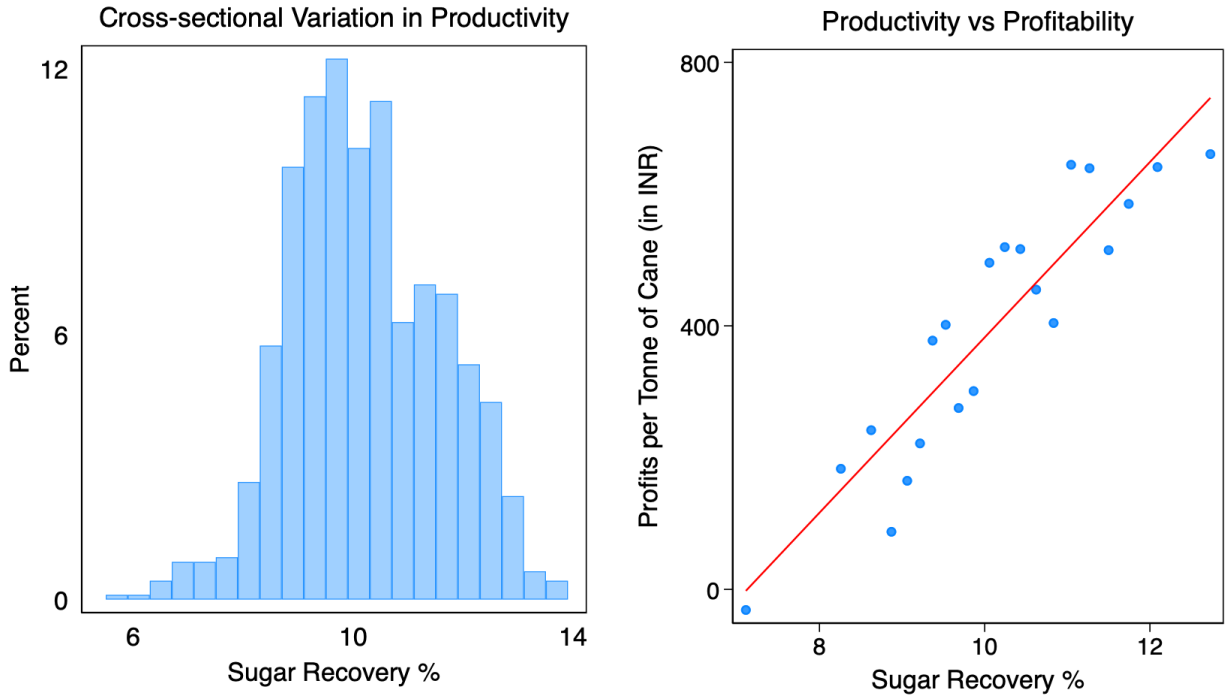
Fact 3: Mills vary widely in productivity, and the least productive operate on the thinnest margins.

I use sugar recovery as an observable measure of mill productivity and find that mills differ enormously in how efficiently they convert cane into sugar. Figure 8, panel A plots the distribution of sugar recovery across mills and reveals substantial cross-sectional dispersion. Recovery ranges from roughly 6 to 14 percent, implying that the least productive mills extract less than half as much sugar per tonne of cane as the most productive. Because sugar revenue is linear in recovery, differences of this magnitude translate into large differences in the value a mill generates from the same quantity of cane. This productivity dispersion maps directly into profitability. Panel B plots profit per tonne of cane against sugar recovery, and shows a strong positive gradient. The least productive mills operate on the thinnest margins, while high-recovery mills earn substantially more per tonne crushed. The relationship is intuitive – a low-recovery mill extracts less sugar from each tonne of cane it purchases, yet pays a cane price anchored by the regulated floor and by competition for cane as described in Fact 2, so its margin is compressed from both sides. Low recovery is therefore not merely a technical inefficiency, it is a direct determinant of financial fragility.

This fragility is what makes productivity dispersion consequential for how the industry responds to drought shocks. Droughts sharply contract cane availability for up to two years after the drought itself. When cane becomes scarce, every mill faces lower throughput, but thin-margin mills have the least buffer to absorb a prolonged period of underutilized crushing capacity. The least productive mills are thus the most exposed to drought. Already operating close to the break-even point, they are the first to become

unviable when a shock decimates local cane supply.

Figure 8: Mills differ in productivity, with laggards facing thinner margins



Panel A: Sugar Recovery

Panel B: Profits

Notes: Panel A shows a cross-sectional histogram on sugar recovery. Panel B shows a binscatter plot for sugar recovery against profits per tonne of cane. *Source:* Plant-level Records from the Indian Directorate of Sugar and the Annual Survey of Industries.

Taken together, the three facts establish the conditions for the paper's mechanism. **Fact 3** identifies which mills are at the margin of survival, and hence which mills a drought is likely to force into exit – a selection that is central to the restructuring mechanism I study. **Fact 1** implies that drought-induced reallocation is not random across space: it removes the least productive mills and reallocates their cane to local survivors, reshaping local market structure and, through the oligopsony power documented in **Fact 2**, the prices farmers ultimately receive. This selection-and-reallocation mechanism is the focus of the model developed in Section 6. The next section presents the event study evidence that these ingredients produce.

5 Empirics

5.1 Empirical Strategy

My identification strategy exploits spatial and temporal variation in the incidence of extreme drought events across India's sugar-producing districts. Extreme droughts are defined as district-year observations where the Standardised Precipitation Evapotranspiration Index (SPEI) falls below -2, corresponding to a 1-in-50 year event. Because sugar production is geographically dispersed across India, different districts experience extreme droughts at different times, providing quasi-experimental variation in the external shock. To analyse the effects of these drought shocks, I implement an event-study design at two levels:

At the **market level**, I use the following specification:

$$y_{dt} = \sum_{k=\underline{T}}^{\bar{T}} \beta_k D_{dt}^k + \delta_d + \lambda_t + \varepsilon_{dt} \tag{1}$$

where y_{dt} is the outcome for district d in year t, δ_d are district fixed effects absorbing time-invariant market characteristics, λ_t are year fixed effects absorbing common annual shocks, and D_{dt}^k are event-time indicators defined relative to the year in which district d experiences its worst extreme drought.

At the **plant level**, I use the following specification:

$$y_{it} = \sum_{k=\underline{T}}^{\bar{T}} \theta_k D_{it}^k + \delta_i + \lambda_t + u_{it} \tag{2}$$

where y_{it} is the outcome for mill i in year t, δ_i are mill fixed effects, λ_t are year fixed effects. Mill-level treatment assignment is based on the drought status of the district in which the mill is located. The coefficient for the event year ($k = 0$) is normalised to zero. Both specifications contain lags and leads from -8 to +12 years around the event respectively, but I restrict the figures to show treatment effects from -5 to +10.

For heterogeneous effects by pre-drought sugar recovery, I interact the event-time indicators with an indicator for whether mill i's lagged recovery falls below the pre-drought

recovery median:

$$y_{it} = \sum_{k=\underline{T}}^{\bar{T}} \theta_k^L D_{it}^k . 1[\omega_{i,\tau-1} < \omega^{med}] + \theta_k^H D_{it}^k . 1[\omega_{i,\tau-1} \geq \omega^{med}] + \delta_i + \lambda_t + u_{it} \tag{3}$$

where ω^{med} is the pre-drought median sugar recovery. The θ_k^L coefficients trace the dynamic response for lower-productivity survivors and θ_k^H for higher-productivity survivors. Standard errors are clustered at the district level throughout.

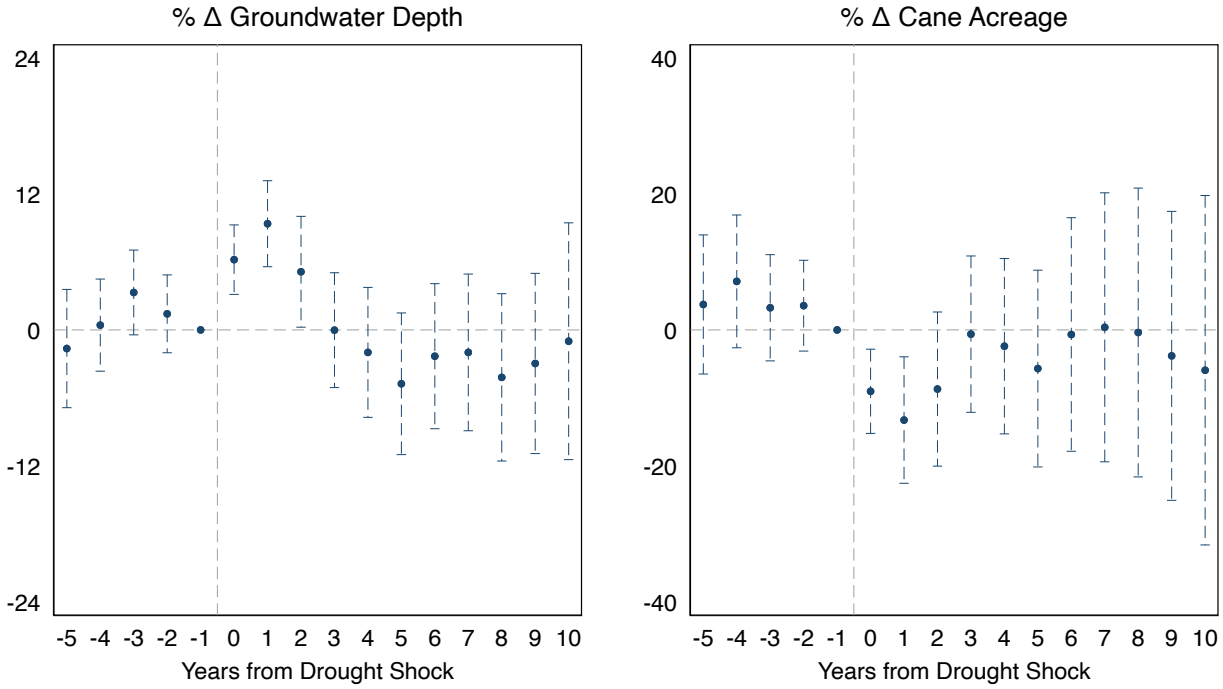
Given the staggered implementation of droughts across districts, I estimate all dynamic treatment effects using the Sun and Abraham (2021) estimator, which addresses concerns about bias from heterogeneous treatment effects in staggered difference-in-differences settings. For robustness, I also estimate using standard two-way fixed effects and show that the results are very similar.

The key identifying assumption is that, in the absence of droughts, outcomes for mills (or districts) in affected areas would have followed similar trends as those in unaffected areas, conditional on mill (or district) and year fixed effects. The features supporting identification are: (i) the incidence of extreme drought is plausibly exogenous to local economic conditions, being driven by large-scale climatic phenomena; (ii) the spatial variation across districts experiencing drought in different years provides multiple treatment cohorts, with nearly a quarter of the districts never experiencing an extreme drought during the sample period, serving as a permanent control group; and (iii) the pre-treatment event-study coefficients are statistically indistinguishable from zero, supporting the parallel trends assumption.

5.2 Results

5.2.1 Result 1: Droughts contract cane supply for 2–3 years

Figure 9: Droughts raise groundwater depths and reduce cane acreage temporarily



Panel A: Log Groundwater Depth **Panel B:** Log Cane Acreage

Notes: Panels A and B present event-study coefficients from Equation 1 using the Sun and Abraham (2021) estimator on annual district-level groundwater depth and cane acreage data respectively. *Source:* Central Groundwater Board (CGWB), ICRISAT and the Indian Directorate of Economics and Statistics.

Extreme droughts have persistent effects on cane supply through a hydrological channel. Figure 9 presents event study estimates of the effect of an extreme drought on market-level groundwater depth and sugarcane acreage. Panel A shows that groundwater depth increases sharply after a drought – by approximately 18 percent in the first two years – and recovers within 3 years as aquifers take time to recharge. Panel B shows the downstream consequence for agriculture: sugarcane acreage and output, which rely on groundwater as their primary source of irrigation, fall significantly in the affected market, with the contraction persisting for nearly 3 years before reverting to pre-drought levels. This fact establishes the shock that triggers the mechanism: an extreme drought generates a multi-year contraction in cane supply that reduces low productivity mills' revenues and potentially pushes them below the viability threshold. Crucially, as I document next,

while the supply shock is temporary, the exit it causes is permanent. The asymmetry between the speed of agricultural recovery and the irreversibility of industrial exit is what generates the persistent shift in market structure and drives the paper's results.

5.2.2 Result 2: Droughts drive the exit of least productive mills, concentrating markets and reallocating exiting mills' farmers to nearby survivors

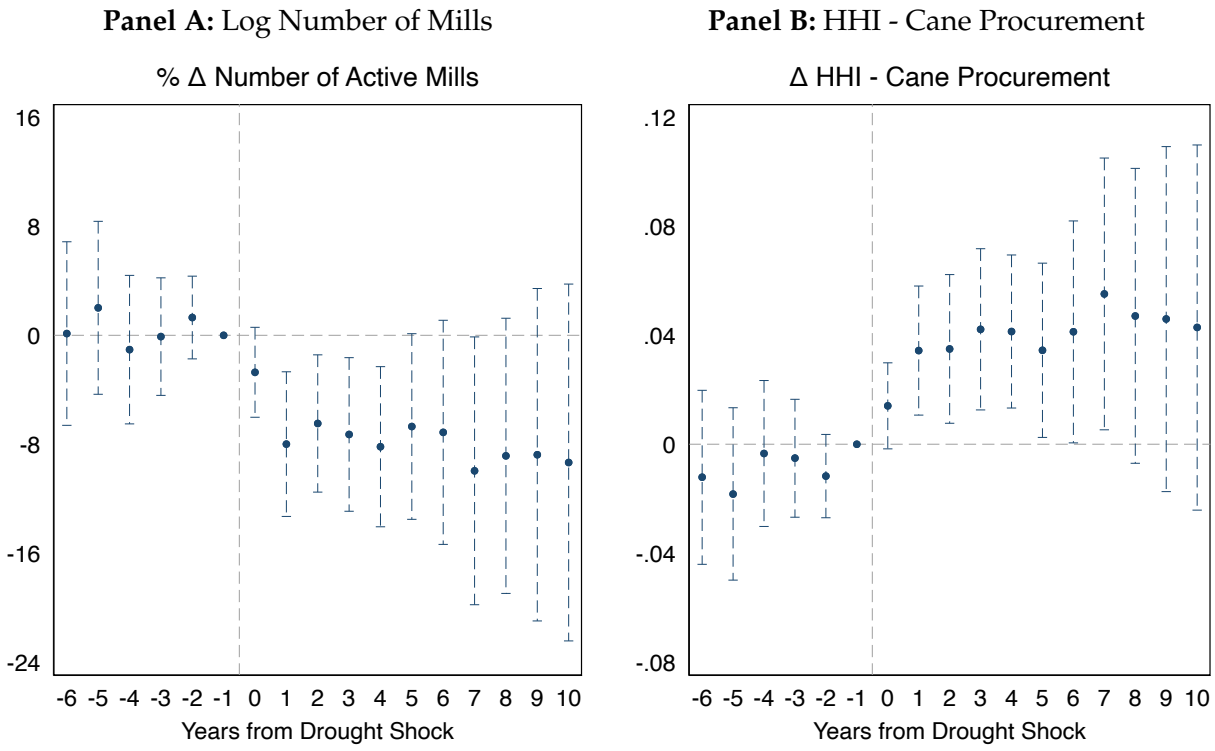


Figure 10: Market concentration sees a persistent rise

Notes: The figure presents event study coefficients from Equation 1 using the Sun and Abraham (2021) estimator. The dependent variables are log number of active mills in Panel A, and the Herfindahl-Hirschman Index (HHI) using cane procurement shares in Panel B. 95% confidence intervals are computed based on standard errors clustered at the district level. *Source:* Plant-level Records from the Indian Directorate of Sugar.

Figure 10, Panel A shows that the number of active mills in a district falls by approximately 10 percent on average in the long run. Panel B shows its consequence for market concentration by plotting a dynamic effect on the Herfindahl-Hirschman Index (HHI) in cane procurement shares. Pre-trends are flat, and after the drought there is a sharp and persistent increase in concentration – the long-run average effect is approximately 0.05. Mills exit and do not re-enter – this is permanent restructuring, not temporary shutdown.

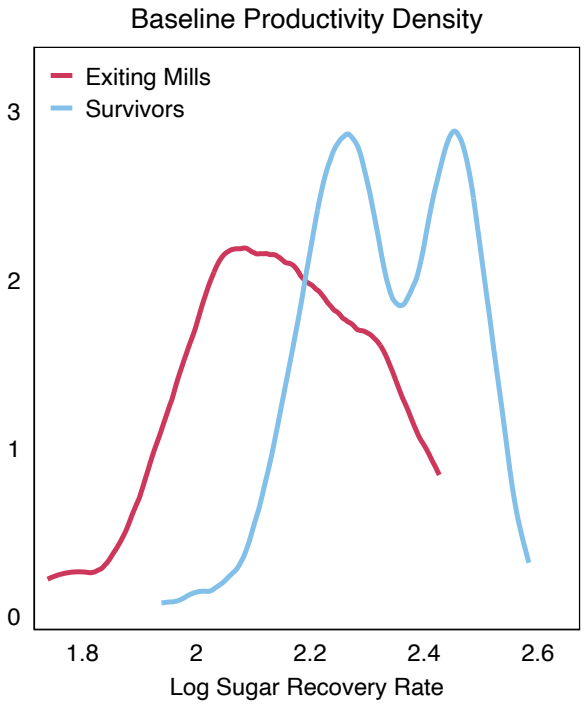
Selection on productivity: Table 1 examines whether drought-induced exit is selective on observed productivity. Column 1 shows that extreme droughts increase the probability of mill exit by 2.9 percentage points. Column 2 interacts drought exposure with an indicator for whether the mill's lagged sugar recovery falls in the bottom tercile of the distribution, where lagged recovery is defined as the mill's average sugar recovery over the two years preceding the drought. Mills in the bottom tercile are 3.6 percentage points more likely to exit on top of the baseline effect of 1.8 percentage points for mills in the top two terciles. Column 3 uses continuous lagged recovery and confirms the same pattern: a one unit increase in lagged sugar recovery reduces the drought-induced exit probability by 1.1 percentage points. Droughts do not kill mills randomly, they selectively remove the least productive ones. The kernel density of log pre-drought sugar recovery in Figure 11 reinforces this finding. Exiting mills are drawn overwhelmingly from the left tail of the productivity distribution at baseline.

Table 1: Plants' exit likelihood rises after droughts, particularly for the least productive

	Likelihood of Exit		
	(1)	(2)	(3)
Drought	0.029^{***}	0.018^{**}	0.126^{***}
	(0.007)	(0.007)	(0.038)
Drought $\times$ **1**{Bottom Tercile of Productivity}		0.036^{***}	
		(0.012)	
Drought $\times$ Lagged Productivity			-0.011^{***}
			(0.004)
Mill FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	5,590	5,590	5,590
Clusters (Districts)	203	203	203

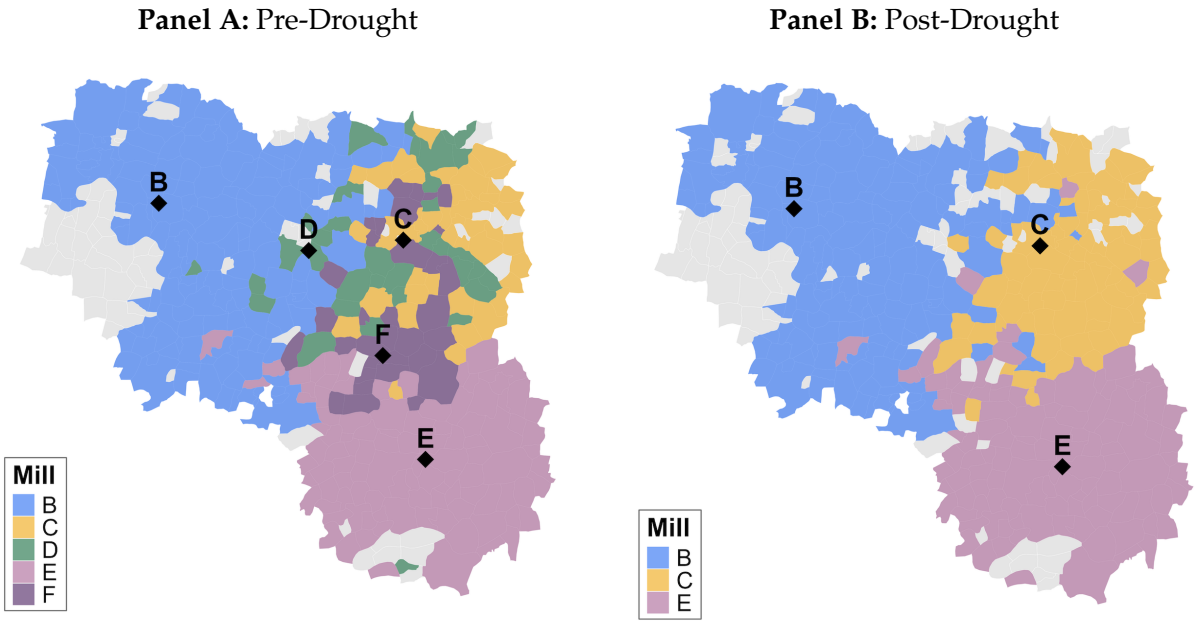
Notes: This table shows plants' likelihood to shut down permanently within 5 years of the drought. The first column shows the coefficient for all plants in my overall sample, while the second column shows the heterogeneity in exit likelihoods based on whether the plant was in the bottom tercile of productivity distribution right before the drought. The third column uses continuous lagged productivity for heterogeneity. Standard errors clustered at the district level in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

Figure 11: Exiting mills are drawn from the left tail of pre-drought recovery distribution



Cane Reallocation: Mill exit drives reallocation of their cane farmers to nearby survivors. Figure 12 provides a vivid illustration from Solapur district in Maharashtra. The left panel maps the pre-drought market structure, with five mills dominating their respective contiguous catchment areas. The right panel shows the post-drought structure: two mills have exited permanently, and their former catchment areas have been absorbed by the three surviving mills, which now control substantially larger spatial footprints. This is the reallocation mechanism in action.

Figure 12: Drought-induced Change in Market Structure in Solapur, Maharashtra



Notes: The figure plots changes in market structure induced by exit of anonymised Mills D and F following a drought in northern Solapur in 2022. *Source:* Plant-level Records from the Indian Directorate of Sugar, and Farmer-Mill Transactions from the Maharashtra Sugar Commissionerate.

5.2.3 Result 3: Lower-productivity survivors are capacity-unconstrained while higher-productivity survivors operate close to full capacity

Before examining how survivors respond to the reallocation of cane from exiting plants, it is important to understand how lower- and higher-productivity survivors differ at baseline. Table 2 compares the two groups – defined by a median split on average sugar recovery over the two years preceding the drought – across a range of operational characteristics measured one year before the drought.

The two groups differ substantially in operational performance. Higher-productivity survivors have significantly higher sugar recovery (10.66% vs 9.14%), crush more cane, operate for a month more during the crushing season, and achieve higher capacity utilisation. They also pay higher cane prices. However, the two groups do not differ in physical scale. Installed crushing capacity is nearly identical at approximately 3,200 tonnes of cane per day (TCD), and mill age is comparable at 28-29 years. The differences between the groups are therefore in how they operate. Lower-productivity survivors have systematically lower capacity utilisation while higher-productivity survivors operate much closer to their capacity constraints.

Table 2: Surviving plants don't differ in scale, but in how they operate

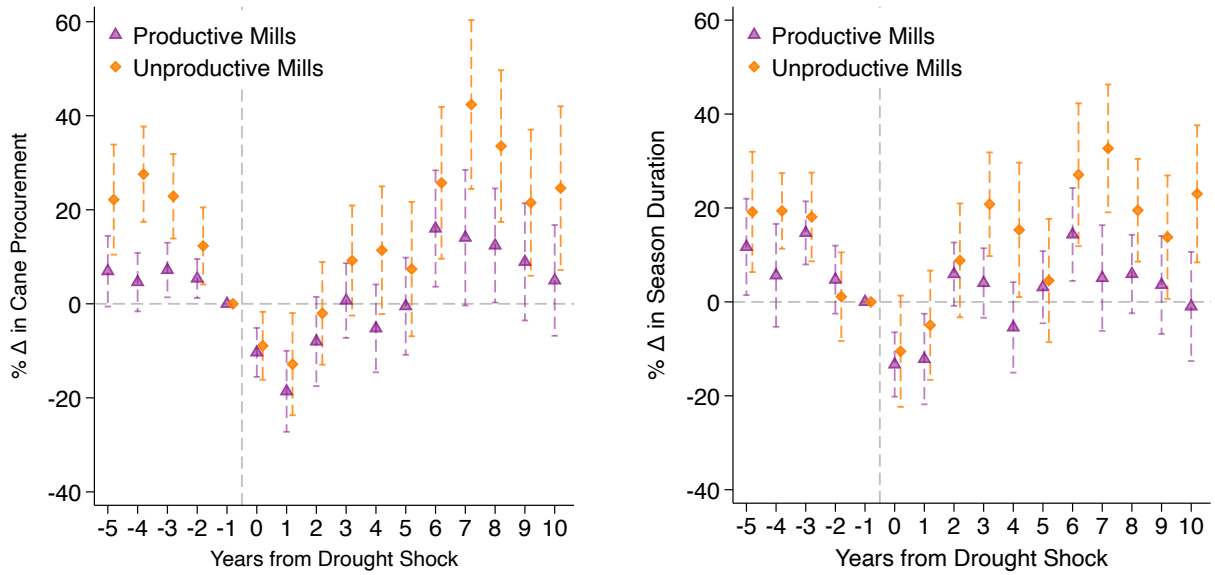
	High Productivity	Low Productivity	$H_a : \Delta \neq 0$
Sugar Recovery (%)	10.66	9.14	0.000
Cane Crushed ('000 tonnes)	475	330	0.000
Crushing Duration (days)	148	121	0.000
Installed Capacity (TCD)	3,300	3,150	**0.350**
Capacity Utilisation (%)	89.4	77.3	0.000
Cooperative	0.41	0.53	0.001
Age (years)	28	29	**0.470**
Permanent Workers	630	514	0.000
Wages ('000 INR)	167	166	**0.970**
Cane Prices (INR/tonne)	1,682	1,515	0.010

Notes: The table reports mean outcomes for high and low productivity surviving plants at baseline before experiencing an extreme drought. High (low) productivity survivors are defined as plants that have operated every year from -5 to +10 years around a drought and have above (below) median sugar recovery over two years preceding the drought. p-values are from two-sided t-tests.

This distinction is central to the reallocation result that follows. When exiting mills' cane is redistributed to surviving mills, lower-productivity survivors can absorb the additional volume because they have spare capacity. Higher-productivity survivors cannot expand their throughput comparably because they are already operating close to the full length of the harvest season as well as their design capacity. Physical capacity expansion which requires adding a new crushing tandem and installing additional boilers requires lumpy capital investment and typically takes 5 years from planning to commissioning, ruling it out as a short-run response. The reallocation therefore flows disproportionately to lower-productivity survivors, not because of any spatial proximity advantage, but because they are the mills with room to accommodate additional cane.

5.2.4 Result 4: Cane from exiting mills is reallocated to survivors, especially to lower-productivity plants

Figure 13: Cane is reallocated primarily to the lower-productivity surviving plants



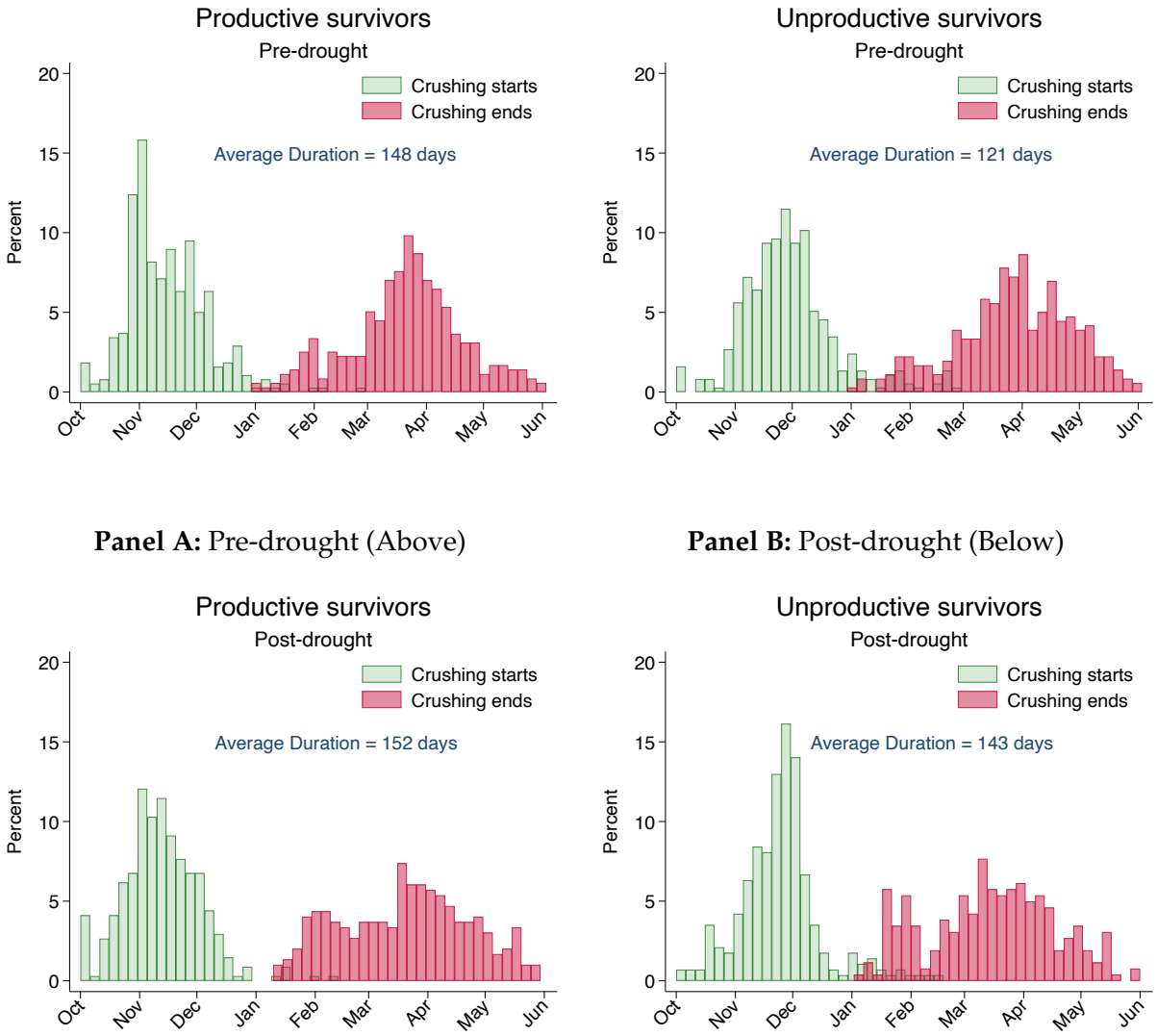
Panel A: Log Cane Procurement

Panel B: Log Season Duration

Notes: The figure presents event study coefficients from Equation 3 using the Sun and Abraham (2021) estimator. The dependent variables are log cane procured in Panel A, and log season duration (in days) in Panel B. 95% confidence intervals are computed based on standard errors clustered at the district level. *Source:* Plant-level Records from the Indian Directorate of Sugar.

Figure 13 presents heterogeneous event-study estimates of the effect of drought on cane procurement and crushing season duration, separately for lower- and higher-productivity survivors. Panel A shows cane procurement: lower-productivity survivors see their procurement expand by 30-40 percent in the years following the drought – a substantial volume windfall. Higher-productivity survivors also gain, but by a much smaller margin of approximately 10 to 15 percent. Panel B presents the results in season duration terms: lower-productivity survivors see their crushing season expand by 30 percent on average, while higher-productivity survivors observe modest gains.

Figure 14: Unproductive survivors observe an expansion in their crushing duration



Panel A: Pre-drought (Above) **Panel B:** Post-drought (Below)

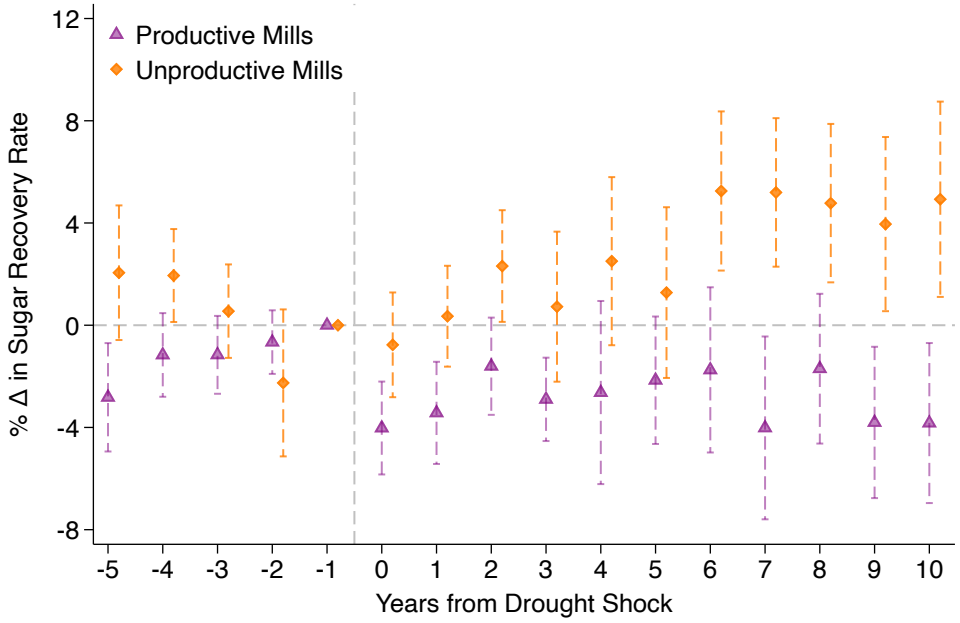
Notes: The figure presents season start and close dates for the full sample of survivors, split into productive and unproductive plants at baseline. Panel A captures this for the pre-drought period, defined as the two years preceding an extreme drought. Panel B captures the same for the post-drought period when cane supply recovers, i.e. years 3 to 5 following the drought. *Source:* Plant-level Records from the Indian Directorate of Sugar.

This is made more evident in Figure 14 which depicts how lower-productivity survivors are able to experience rapid expansion in crushing operations relative to their higher-productivity counterparts. The former begin their crushing operations three weeks earlier, on average, after cane supply reverts to normal levels following extreme droughts. Given the harvesting season typically lasts for 5 months from November to April, and

higher-productivity survivors are already operating throughout this duration, it's the lower-productivity mills who absorb the vast majority of cane that is reallocated from exiting plants. As such, this disproportionate reallocation is a direct consequence of the capacity constraint asymmetry also documented in Table 2. It is this volume windfall – exogenous to the mill's own investment decisions – that raises the marginal return to quality upgrading and triggers the within-firm productivity gains documented in the following subsection.

5.2.5 Result 5: Productivity gains are concentrated among lower-productivity survivors, and realised through management upgrading

Figure 15: Sugar recovery rises persistently for lower-productivity survivors

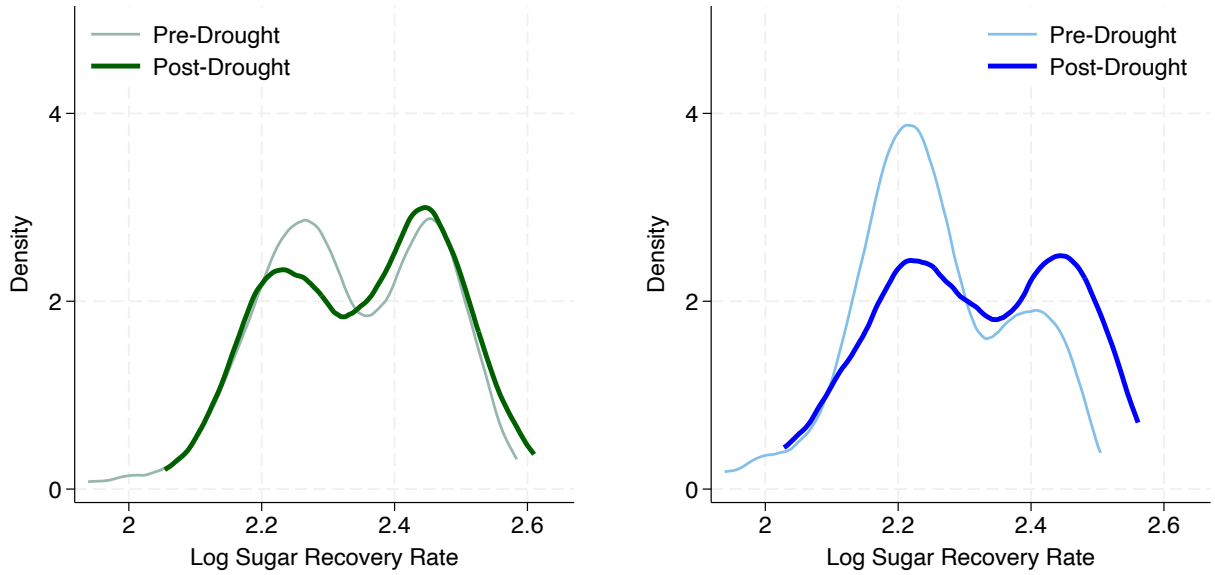


Notes: The figure presents event study coefficients from Equation 3 using the Sun and Abraham (2021) estimator. The dependent variable is log sugar recovery rate. 95% confidence intervals are computed based on standard errors clustered at the district level.

Sugar Recovery: Figure 15 presents the paper's key result. Sugar recovery – the primary measure of mill productivity – rises significantly for lower-productivity survivors but not for higher-productivity ones. The lower-productivity survivors show gains of approximately 6 percent by 5-10 years after the drought, whereas effects remain flat for higher-productivity plants over the same horizon. Figure 16 confirms this using kernel density estimates of the pre- and post-drought recovery distribution. In panel A which shows all surviving mills, the distribution has shifted rightward 8 to 10 years after the

drought. When isolating the effect for lower-productivity survivors in panel B, the shift appears much larger and represents significant within-firm convergence.

Figure 16: Sugar recovery gains are driven by lower-productivity survivors

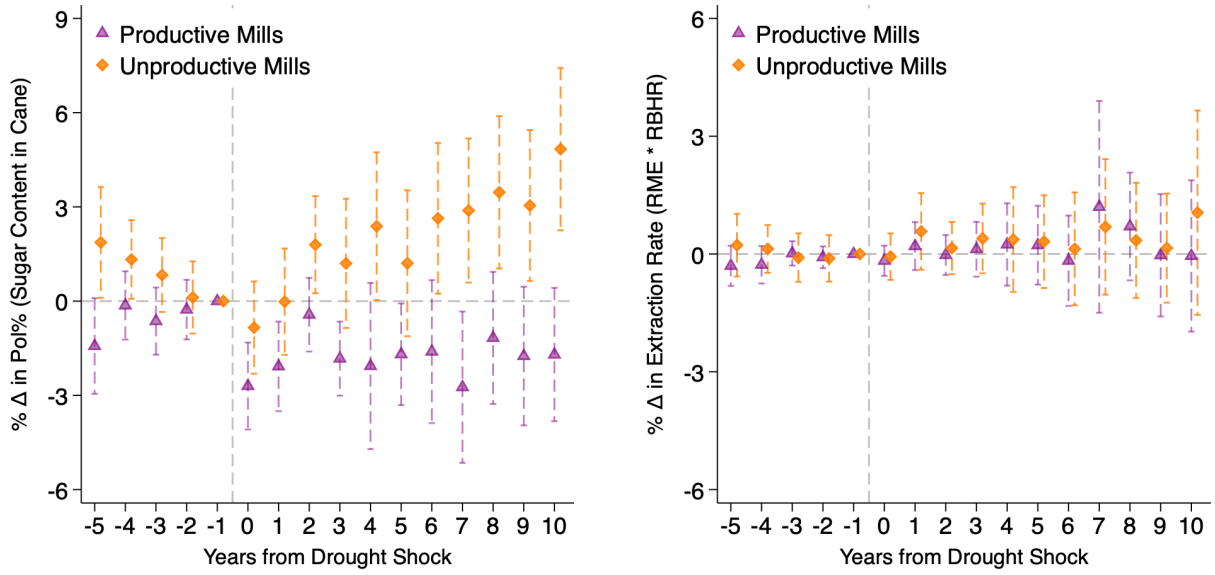


Panel A: Full Sample of Survivors **Panel B:** Lower Productivity Survivors

Notes: The figure plots kernel density estimates of log sugar recovery rates for the full sample of surviving plants in Panel A, and for lower-productivity surviving plants in Panel B. Pre-drought densities are based on mean plant recovery rates in the two years before a drought, and post-drought densities are based on mean plant recovery rates between 8-10 years following the drought. *Source:* Plant-level Records from the Indian Directorate of Sugar.

Decomposition – Sugar Content versus Extraction Losses: Fact 4 demonstrated that sugar recovery is determined by two components: the sugar content in the cane (driven by cane maturity and varietal composition) and extraction losses (driven by processing efficiency). Figure 17 decomposes the recovery gains into these two components. Panel A shows effects on Pol%, a direct measure of sugar content in cane, which rises sharply for lower-productivity survivors, by approximately 3 to 5 percent in the medium to long run, while remaining flat for higher-productivity survivors. Panel B shows the extraction rate – measured as the product of Reduced Mill Extraction and Reduced Boiling House Recovery, both of which are explicitly constructed to control for cane quality – and there is no differential effect. The entire productivity gain thus comes through the sugar content channel, not through improvements in processing technology.

Figure 17: Productivity gains are realised through cane quality management



Panel A: Sugar Content in Cane

Panel B: Extraction Rate

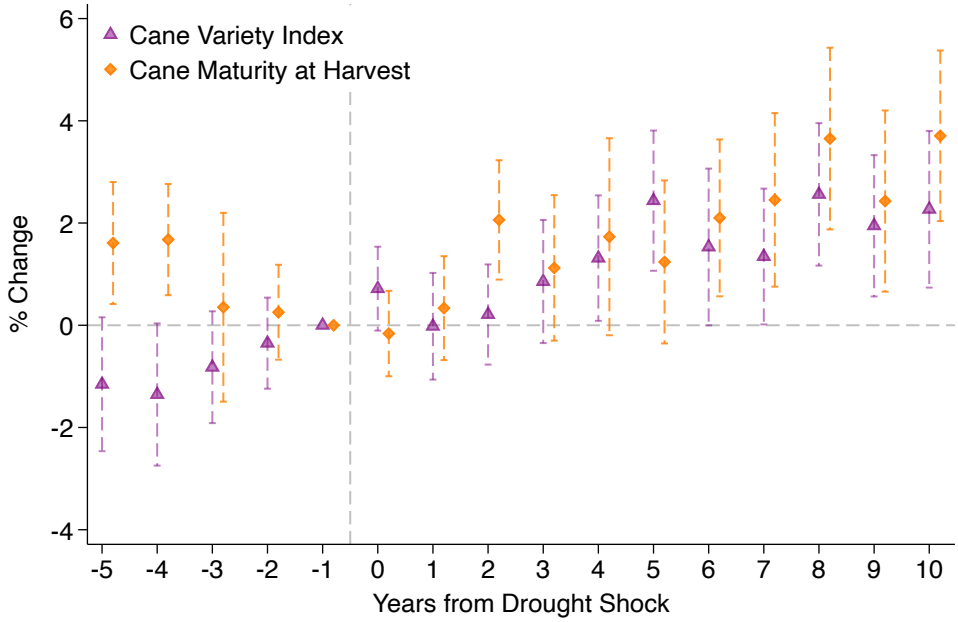
Notes: The figure presents event study coefficients from Equation 3 using the Sun and Abraham (2021) estimator. The dependent variables are log Pol% (sugar content) in Panel A, and log extraction rate in Panel B. 95% confidence intervals are computed based on standard errors clustered at the district level. *Source:* Plant-level Records from the Indian Directorate of Sugar.

This decomposition has important implications for the model. Improving sugar content in cane is fundamentally a management activity: it requires coordinating with thousands of farmers on harvesting schedules, distributing improved cane varieties, and deploying maturity measurement in the field, rather than purchasing new machinery or upgrading factory equipment. The null result on extraction losses, measured by industry-standard engineering metrics that are specifically designed to be orthogonal to cane quality, rules out the processing technology channel and validates the model's focus on the input quality management channel.

Management Upgrading: Figure 18 provides direct evidence of the management upgrading channel. For lower-productivity survivors, both measures of management practice improve significantly after drought. The Cane Variety Index – a measure of average cane quality based on sucrose content of procured varieties – rises by approximately 3 percent six-to-ten years after the drought. Brix% in cane primary juice – a measure of whether cane is being harvested at optimal maturity – shows 4 percent gains over the same horizon. While improved cane varieties possess higher sucrose content, optimising cane harvesting operations helps accumulate more sucrose as the cane matures (McDon-

ald and Lisson, 2001; Khandagave and Patil, 2007; Singh et al., 2026) – directly impacting sugar content in cane.

Figure 18: Improved cane varieties and maturity at harvest drive productivity gains



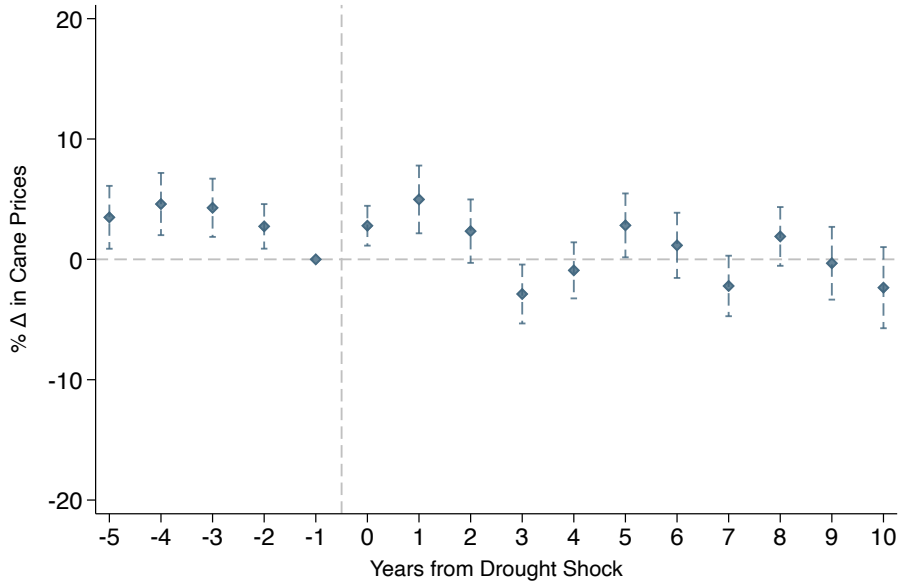
Notes: The figure presents event study coefficients from Equation 2 using the Sun and Abraham (2021) estimator for lower-productivity survivors. The dependent variables are log Brix% (cane maturity) and cane variety index. 95% confidence intervals are computed based on standard errors clustered at the district level. *Source:* Plant-level Records from the Indian Directorate of Sugar.

Both mechanisms reflect investments in the mill's *management* of its cane supply chain, consistent with the interpretation that upgrading operates through improved management practices rather than capital investment in processing technology. This aligns with the broader literature on the importance of management quality for firm productivity in developing countries (Bloom et al., 2013), while identifying a specific channel through which management improvement is triggered by exogenous changes in market structure.

5.2.6 Result 6: Strikingly, farmers see no corresponding rise in cane prices

Figure 19 presents event-study estimates of the effect of drought on farm-gate cane prices for lower-productivity survivors. Despite the substantial productivity gains documented above, cane prices show no corresponding increase. The point estimates are positive immediately after the drought reflecting demand when cane supply has contracted owing to drought. Prices, however, hover around zero and, if anything, are slightly negative in the medium-to-long run.

Figure 19: Cane prices do not rise in response to within-firm productivity gains



Notes: The figure presents event study coefficients from Equation 2 using the Sun and Abraham (2021) estimator for lower-productivity survivors. The dependent variable is log cane price. 95% confidence intervals are computed based on standard errors clustered at the district level. *Source:* Plant-level Records from the Indian Directorate of Sugar and the Annual Survey of Industries.

This result is striking: sugar recovery for lower-productivity survivors has risen by 6 percent, implying that the marginal revenue product of their cane has increased proportionally. If cane markets were competitive, these gains would pass through to farmers as higher cane prices. The fact that prices are flat while productivity rises implies that oligopsony markdowns – the gap between the marginal revenue product of cane and the price paid to farmers – have widened. The productivity gains from within-firm upgrading appear to be captured almost entirely by mills rather than shared with their upstream suppliers. This is the distributional consequence of climate-induced restructuring: aggregate productivity rises, but the gains accrue to processors at the expense of farmers.

5.3 Summary of Empirical Results

The reduced-form evidence establishes five results that, taken together, trace the causal chain from drought-induced restructuring to within-firm upgrading:

1. Selective exit of least productive plants raises market concentration.
2. Surviving plants differ in operational performance, not physical scale.

3. Cane from exiting plants is reallocated disproportionately to lower-productivity, capacity-unconstrained survivors.
4. These lower-productivity survivors exhibit large within-firm productivity gains, realised by management upgrading
5. Cane prices do not rise, despite within-firm gains.

These five results motivate and discipline the structural model developed in the next section. The model must rationalise why lower-productivity survivors, who receive the largest volume windfall and are furthest from the productivity frontier, exhibit the strongest upgrading response. It must also explain why these productivity gains are not passed through to farmers, and quantify the welfare trade-off between aggregate productivity improvement and distributional cost.

6 Model

6.1 Intermediary Cane Demand

6.1.1 Technology

Sugar mill i in year t produces sugar Q_{it} using cane M_{it}, labour L_{it}, which are both expressed in quantities, and fixed assets K_{it}. I assume that cane cannot be substituted with either labour or capital, so production is Leontief:

$$Q_{it} = \min \left\{ \beta_{it}^{M} M_{it}, \;\; \Omega_{it} H(L_{it}, K_{it}) \right\}. \tag{4}$$

The first arm corresponds to the amount of sugar recovered directly from cane: a tonne of cane yields β_{it}^{M} tonnes of sugar, where β_{it}^{M} is *sugar recovery*, the observable measure of mill productivity. The second arm is a labour-capital aggregate $H(\cdot)$, twice differentiable in both inputs with common parameterisation β, scaled by Hicks-neutral productivity Ω_{it}. Processing capacity is bounded: cumulative crushing cannot exceed $\bar{M}_{it}$, the mill's maximum seasonal throughput, which I construct as the mill's installed crushing capacity (tonnes crushed per day) multiplied by the length of the crushing season in its state (days), inferred from manufacturing records:

$$M_{it} \;\leq\; \bar{M}_{it} \;\equiv\; \underbrace{\kappa_{it}}_{\text{Installed Capacity (TCD)}} \times \underbrace{D_{s(i)t}}_{\substack{\text{Season Length} \\ (days)}} \tag{5}$$

Departing from the common assumption that the conversion rate is homogeneous across firms (Rubens, 2023; Avignon and Guigue, 2025) or exogenous (De Loecker and Scott, 2016), I treat recovery as a plant-level *choice* which can be improved through cane-quality upgrading, motivated by the empirical results shown in Section 5. Upgrading is a discrete management decision $u_{it} \in \{0,1\}$: a mill that adopts the cane-quality upgrading programme raises recovery by a factor $\gamma > 1$, at a fixed cost F_{up},

$$\beta_{it}^{M}(u_{it}) = \left(1 + (\gamma - 1)\, u_{it}\right) \beta_{it}^{M} \;=\; \begin{cases} \beta_{it}^{M} & \text{operate as normal } (u_{it} = 0), \\ \gamma\, \beta_{it}^{M} & \text{upgrade } (u_{it} = 1). \end{cases} \tag{6}$$

Treating the conversion rate as heterogeneous and firm-chosen is the key institutional feature of agricultural processing, not only in sugar but in coffee, cotton, tea, and other goods, something that a homogeneous-recovery model omits. Equation 4 nests production functions in which all inputs are substitutable; the recovery parameter would then be zero, and cane would be an additional substitutable input in $H(\cdot)$.

I assume that cane and labour are variable inputs: their amount is chosen by mills in the same time period as when they are used. I also assume that cane and labour are static inputs, which means that they only affect current profits. These assumptions rule out adjustment costs over sugar and inventories. Capital is, in contrast, dynamic and fixed; the capital stock at time t can only be changed through investment at time t - 1.

6.1.2 The Mill's Problem

Each year, mills choose labour L_{it}, cane M_{it}, and the discrete upgrading decision $u_{it} \in \{0,1\}$ to maximise current variable profits. Because labour and cane are complements, choosing labour pins down cane and hence output. Through their cane quantity choices, mills also set cane prices subject to a floor. Hence, cane farmers are assumed to be price-takers, which is intuitive given that there are many more farmers than sugar mills ($\sim$15,000 farmers per mill). Mills are price-takers in sugar and labour (P_t^s, W_{it}^L) and oligopsonists in cane.

$$\Pi_{it} = \max_{L_{it},\, M_{it}} \max \Big\{ \underbrace{P_t^s \beta_{it}^M M_{it} - W_{it}^M M_{it} - W_{it}^L L_{it} - F_{\text{op}}}_{\text{operate as normal}},\; \underbrace{P_t^s \gamma \beta_{it}^M M_{it} - W_{it}^M M_{it} - W_{it}^L L_{it} - F_{\text{op}} - F_{\text{up}}}_{\text{upgrade } (\gamma>1,\text{ pay } F_{\text{up}})} \Big\}, \tag{7}$$

where sugar revenue uses $Q_{it} = \beta_{it}^M M_{it}$ from (4), and F_{op} is a fixed operating cost. The problem is subject to the capacity constraint and the statutory price floor, which has base

price ϕ_0 at reference recovery $\bar{\beta}^M$ and a recovery premium $\phi_1 > 0$:

$$M_{it} \;\le\; \bar{M}_i, \qquad W^M_{it} \;\ge\; \phi_0 + \phi_1\big(\tilde{\beta}^M_{it} - \bar{\beta}^M\big), \tag{8}$$

where $\tilde{\beta}^M_{it} \in \{\beta^M_{it}, \gamma\beta^M_{it}\}$ is the mill's realised recovery. The floor is the Fair and Remunerative Price (or, in SAP states, the higher State Advised Price); its recovery-linked premium means an upgrading mill faces a higher mandated price. A mill operates only if its variable profit is non-negative; otherwise it exits, $\pi(\beta^M_{it}) < 0 \Rightarrow$ exit, withdrawing its cane demand and, as shown below, widening the markdowns of surviving rivals.

6.1.3 Input Prices and Markdowns

Solving the cane first-order condition delivers the cane-price markdown:

$$W^M_{it} = \frac{\varepsilon^s_{it}}{\varepsilon^s_{it}+1}\Big(\mathrm{NetMRP}^M_{it} - \lambda_{it}\Big), \qquad \mathrm{NetMRP}^M_{it} \equiv \tilde{\beta}^M_{it}\Big(P^s_t - \tfrac{W^L_{it}}{\Omega_{it}H_L}\Big), \tag{9}$$

where NetMRP^M_{it} is the marginal revenue product of cane net of the complementary labour cost of processing it, $\lambda_{it} \ge 0$ is the shadow value of the capacity constraint with $\lambda_{it}(\bar{M}_i - M_{it}) = 0$, and $H_L \equiv \partial H/\partial L$. The factor $\varepsilon^s_{it}/(\varepsilon^s_{it}+1) < 1$ is the oligopsony markdown: a less elastic residual cane supply (lower ε^s_{it}) widens the markdown – for instance when drought-induced exit concentrates procurement on a surviving mill. The capacity multiplier depresses the price further: a mill at its crushing limit values marginal cane only up to its processing value, so $\lambda_{it} > 0$ lowers W^M_{it}. When the floor binds the mill instead pays $W^M_{it} = \phi_0 + \phi_1(\tilde{\beta}^M_{it} - \bar{\beta}^M)$.

6.1.4 The Upgrading Decision

Upgrading is discrete: a mill upgrades iff the upgrade payoff in (7) weakly exceeds the operate-normally payoff. When the floor is slack the mill pays the markdown $W^M_{it} = \psi_{it}P^s_t\tilde{\beta}^M_{it}$ with $\psi_{it} \equiv \varepsilon^s_{it}/(\varepsilon^s_{it}+1)$, keeping only the $(1-\psi_{it})$ share of the value it creates; the labour and operating terms cancel and the decision reduces to

$$(1-\psi_{it})\,P^s_t\,(\gamma-1)\,\beta^M_{it}\,M_{it} \;\ge\; F_{\mathrm{up}} \quad\Longleftrightarrow\quad M_{it} \;\ge\; M^*_{it} \equiv \frac{F_{\mathrm{up}}}{(1-\psi_{it})\,P^s_t\,(\gamma-1)\,\beta^M_{it}} \tag{10}$$

Two implications follow.

(i) **Scale-Upgrading Complementarity.** The upgrade benefit is increasing in cane volume, so only mills above the threshold M^*_{it} upgrade. Reallocation of cane towards

survivors raises their scale and pushes them past the threshold, so absorbing mills upgrade and aggregate recovery rises with concentration.

(ii) **Buyer Power.** The benefit also scales with the captured share $(1 - \psi_{it})$, which is larger when residual supply is less elastic. Greater buyer power lowers M^*_{it}, so the same restructuring that raises a survivor's scale also raises the share of the upgrading gain it retains.

Market restructuring thus raises both the incentive to upgrade (scale M) and the power to capture the gains (the markdown share $1 - \psi$), pushing survivors past M^*.

6.1.5 Cane Input Demand

Inverting the labour condition under a slack price floor and slack capacity gives optimal labour, and substituting into (4) delivers the cane input-demand function which is truncated at $\bar{M}_i$ when capacity binds.

$$M^*_{it} = \frac{1}{\tilde{\beta}^M_{it}} \left(\Omega_{it} \left(\beta^L\right)^{\beta^L} K_{it}^{\beta^K} \left[\frac{P^s_t - \frac{W^M_{it}}{\tilde{\beta}^M_{it}} \left(\frac{\varepsilon^s_{it}+1}{\varepsilon^s_{it}} \right)}{W^L_{it}} \right]^{\beta^L} \right)^{\frac{1}{1-\beta^L}} \tag{11}$$

Compactly, $M_{it} = m\left(P^s_t, W^M_{it}, W^L_{it}, K_{it}, \beta^L, \beta^K, \tilde{\beta}^M_{it}, \Omega_{it}, \varepsilon^s_{it}\right)$, which is strictly monotone in productivity Ω_{it}. Except the production parameters $(\beta^L, \beta^K, \Omega_{it})$, all other parameters on the right-hand side $(P^s_t, W^M_{it}, W^L_{it}, L_{it}, K_{it}, \tilde{\beta}^M_{it})$ are observed, with ε^s_{it} coming from the upstream supply model. Monotonicity in Ω_{it} makes (11) the proxy that inverts for latent productivity in estimation using the control function approach of Ackerberg, Caves and Frazer (2015).

6.2 Upstream Supply

Farmers face a nested discrete choice problem. In the inner nest, a farmer who has chosen to grow cane selects which mill to sell to. In the outer nest, the farmer decides whether to grow cane at all, or to allocate land to an alternative crop. The nesting structure ensures that mill exit affects farmers through two channels: it reduces the set of mills available (inner nest) and, through the inclusive value, reduces the attractiveness of cane relative to other crops (outer nest).

6.2.1 Environment

A market d (a district-year) contains a unit mass of land plots indexed by ω, with cultivable area $\bar{L}_d$. Each plot is operated by a farmer who allocates its use to the crop with the highest revenue per hectare. The choice set has two levels. At the outer level the plot is allocated to one of $K+1$ crops $k \in \{0, 1, \ldots, K\}$, where $k = 0$ denotes sugarcane and $k = K$ a reference crop. At the inner level, conditional on cane, the plot's output is sold to one of the mills i from the local choice set $I_d \cup \{0\}$, where $\{0\}$ denotes an outside option (selling to a jaggery producer).

For a competing crop $k \geq 1$, revenue per hectare is $\mathcal{R}_k(\omega) = \tilde{p}_k \, \eta_k(\omega)$, where $\tilde{p}_k$ is the crop's effective output price and $\eta_k(\omega)$ is the plot's idiosyncratic productivity. For cane sold to mill i, revenue is $\mathcal{R}_{0i}(\omega) = P_i \, \eta_0(\omega) \, \nu_i(\omega)$, where η_0 is cane productivity, ν_i a plot–mill match shock, and P_i the value of mill i per efficiency unit of cane. I parameterize the log of this value as linear in observables,

$$\ln P_i \;\equiv\; \tfrac{1}{\theta}\,\delta_i, \qquad \delta_i(d,t) = \alpha \ln W_i^M(d,t) + \mathbf{x}_{it}'\boldsymbol{\gamma} + \xi_i(d,t), \qquad \delta_0 \equiv 0, \tag{12}$$

where W_i^M is the cane price offered by mill i, and $\mathbf{x}_{it}$ captures observable mill characteristics. The coefficient $\alpha > 0$ governs the price response, and the term $\xi_i(d,t)$ captures unobserved mill-specific amenity.

Nesting assumption. Competing crop productivities are independent Fréchet, $\Pr[\eta_k \leq \eta] = \exp(-T_k \eta^{-\theta})$, with shape $\theta > 1$ and scale $T_k > 0$, independent across crops and plots. The cane revenues $\{\mathcal{R}_{0i}(\omega)\}_i$ are instead positively dependent within the cane nest: they share the common cane component $\eta_0(\omega)$, while the mill match shocks $\nu_i(\omega)$ are i.i.d. across mills. The strength of this within-nest dependence is governed by a dissimilarity parameter $\rho_0 \in (0, 1]$ ($\rho_0 = 1$ being independence). I take this correlation as a primitive, built into the joint distribution of cane revenues rather than derived by adding shocks, consistent with what a nested GEV generator provides (McFadden, 1977). In logs, $\ln \mathcal{R}_{0i}$ is a deterministic term, an i.i.d. Gumbel mill shock at scale σ, and a common cane-nest component – a Cardell(ρ_0) variate (Cardell, 1997) – that ties the mills together. Since the log of a Fréchet variate is Gumbel, revenue maximization in levels is equivalent to additive random-utility maximization in logs, so the Fréchet outer nest and the Gumbel inner nest are two levels of a single nested-GEV problem. I exploit this equivalence throughout: it delivers closed-form shares and, as shown below, a testable cross-parameter restriction.

6.2.2 Inner Nest: Mill Choice Conditional on Growing Cane

Under the nesting assumption above, the probability that a farmer sells to mill i is:

$$s_{i|0}(d,t) = \frac{\exp(\delta_i/\sigma)}{\sum_{i' \in \mathcal{I}_d \cup \{0\}} \exp(\delta_{i'}/\sigma)} \tag{13}$$

where $\sigma > 0$ is the logit scale of the mill nest. The inclusive value of the cane nest summarises the expected payoff from the best available mill option:

$$IV_0(d,t) = \sigma \ln \left(\sum_{i \in \mathcal{I}_d \cup \{0\}} \exp(\delta_i/\sigma) \right) \tag{14}$$

This object has three properties that matter for the model. First, IV_0 is increasing in the number and quality of available mills: adding a mill strictly raises IV_0, and removing one (through exit) strictly lowers it. Second, IV_0 is increasing in any mill's cane price, since higher prices raise δ_i. Third, IV_0 links the inner nest to the outer nest: it enters the crop choice as the effective cane price, so that changes in the mill landscape - exit, entry, or price changes - propagate to farmers' crop allocation decisions.

6.2.3 Outer Nest: Crop Choice on Heterogeneous Land

In the outer nest, the farmer allocates land to a crop $k \in \{0 \text{ (cane)}, 1, \ldots, K\}$ with the highest revenue. Following Costinot, Donaldson and Smith (2016), each plot has a vector of crop-specific productivities drawn from independent Fréchet distributions with location parameters $T_k(d)$ (capturing land suitability for crop k in market d) and common shape parameter $\theta > 1$ (governing the dispersion of comparative advantage across plots). The share of land allocated to crop k is:

$$s_k(d,t) = \frac{T_k(d) \cdot \tilde{p}_k(d,t)^\theta}{\sum_{k'} T_{k'}(d) \cdot \tilde{p}_{k'}(d,t)^\theta} \tag{15}$$

where $\tilde{p}_k$ is the effective price of crop k. For cane, the effective price is derived from the inclusive value: $\tilde{p}_0(d,t) = \exp(IV_0(d,t)/\alpha)$, which aggregates over all mill options weighted by their attractiveness. For alternative crops, $\tilde{p}_k$ is the market price.

The Fréchet structure captures heterogeneity in land suitability: some plots are well suited to cane (e.g., irrigated, alluvial soils) while others are better for soybean or pulses. The shape parameter θ governs the aggregate acreage response to price changes: higher θ

means more plots are on the margin between cane and alternative crops, so a given price change induces a larger shift in acreage.

The cane supply in market d to mill i is:

$$C_i(d,t) = \bar{L}_d \cdot y_0(d,t) \cdot s_0(d,t) \cdot s_{i|0}(d,t) \tag{16}$$

where $\bar{L}_d$ is the cultivable land area, y_0 is yield per hectare, s_0 is the cane acreage share (from the outer nest), and $s_{i|0}$ is the within-cane mill share (from the inner nest).

Consistency of the two nests imposes a restriction that is central to interpretation and identification. Requiring (13) and (15) to arise from a single valid nested-GEV generator pins the dissimilarity to

$$\rho_0 = \frac{\theta\sigma}{\alpha} \in (0,1], \qquad \text{equivalently} \qquad \frac{\alpha}{\sigma} \geq \theta. \tag{17}$$

Economically, mills must be closer substitutes for one another than cane is for competing crops – the within-cane substitution elasticity α/σ weakly exceeds the across-crop elasticity θ. I do not impose (17); I estimate its two sides separately and treat it as an overidentifying restriction on the nesting structure.[15]

6.2.4 Supply Elasticity

The own-price supply elasticity faced by mill i combines two margins of farmer response:

$$\varepsilon_i^s = \underbrace{\frac{\alpha}{\sigma}(1 - s_{i|0})}_{\substack{\text{Within-cane margin:} \\ \text{steal farmers from rival mills}}} + \underbrace{\theta\, s_{i|0}(1 - s_0)}_{\substack{\text{Cross-crop margin:} \\ \text{pull farmers from other crops into cane}}} \tag{18}$$

The within-cane margin captures substitution among mills: when mill i raises its price, it attracts farmers from competing mills, with the response governed by α/σ and proportional to $(1 - s_{i|0})$ - the share of cane farmers currently selling to other mills. The cross-crop margin captures substitution between cane and alternative crops: a higher cane price raises IV_0, pulling some plots from soybean or cotton into cane, with the response proportional to $s_{i|0}(1 - s_0)$.

The elasticity has a key comparative static for the paper's mechanism. When competi-

[15] A proof that $\rho_0 \leq 1$ is necessary and sufficient for a valid GEV generator, via McFadden (1977), is given in the appendix.

tors exit following a drought, each surviving mill's within-cane share $s_{i|0}$ rises mechanically (fewer mills sharing the same farmers). This lowers ε_i^s - the mill faces a less elastic residual supply - increasing its oligopsony power. This is the market power channel of drought-induced restructuring: exit raises concentration and widens the markdown that each surviving mill can extract.

6.3 Equilibrium

Given the regulated sugar price P^s, labour wage w, fixed costs F_{op} and F_{up}, and the cane-price floor $\underline{W}^M$, a *market equilibrium* is a cane-price vector $\{W_i^M\}$, a set of mill decisions, and a set of farmer allocations such that:

1. **Farmers optimize.** Taking $\{W_i^M\}$ as given, each farmer chooses a crop (outer nest) and, conditional on cane, a mill (inner nest). These choices determine the residual cane supply C_i and the supply elasticity ε_i^s faced by each mill i.
2. **Mills optimize.** Taking its residual supply as given, each mill i chooses labour, cane, whether to upgrade, and whether to exit, and sets the markdown price $W_i^M \geq \underline{W}^M$.
3. **The cane market clears.** Cane supply C_d equals cane demand M_d at $\{W_i^M\}$ in each market. Mills are price-takers in the sugar and labour markets, so only the cane market clears.

Since each W_i^M enters both mill i's demand and, through $\{\delta_{i'}\}$, all farmers' shares, 1-3 constitute a fixed point in $\{W_i^M\}$. I solve it by iterating on the price vector, exploiting the monotonicity of excess cane demand.

7 Estimation

7.1 Production Function

Taking the logarithm of the production function in (4) results in equation (19). As cane is assumed to be non-substitutable and a linear function of the sugar output, it does not enter the estimable production function. The production function coefficients β need to be identified:

$$q_{it} = h(l_{it}, k_{it}, \beta) + \omega_{it}. \tag{19}$$

The "output price bias" of Klette and Griliches (1996) is taken into account because output is observed in physical units rather than as revenues. Labour is observed in physical units

as well but potentially with error; rather than observing total hours worked of quality-adjusted labour l_{it}, I observe the number of employees without quality adjustment, $\tilde{l}_{it}$. Capital is measured in monetary values $\tilde{k}_{it}$. If latent differences in utilization of labour are correlated with mill productivity, this induces an "input price bias" as discussed in De Loecker et al. (2016). To address this concern, I add a function $a(\cdot)$ of wages per worker to the production function:

$$q_{it} = h(\tilde{l}_{it}, k_{it}, \beta) + a(w_{it}^{L}) + \omega_{it}. \tag{20}$$

Identification. In order to identify the production function, I impose timing assumptions on mills' input choices, as proposed by Olley and Pakes (1996). As was explained in Section 6.2.1, labour is assumed to be a variable and static input, whereas capital is fixed and dynamic. Hence, labour is assumed to be chosen at time t, after the productivity shock ω_{it} is observed by the mill, while capital investment is chosen at time t–1, before the productivity shock is observed.

The usual approach in the literature is to invert the intermediate input demand function to recover the latent productivity level ω_{it}, which can be used to construct the innovation using a productivity law of motion (Levinsohn and Petrin, 2003; Ackerberg, Caves and Frazer, 2015). This approach hinges on productivity being the only latent, serially correlated input demand shifter. However, input demand varies due to markdown variation as well, as pointed out by Doraszelski and Jaumandreu (2021). This becomes a key concern in my setting given drought shocks lead to concentrated markets following exit of less productive mills, potentially altering cane price markdowns observed by surviving mills. The approach with input inversion can still be used when making additional parametric assumptions on the distribution of markdowns, as I do in my nested supply model. Alternatively, one can also impose more structure on the productivity transition process, with the benefit of allowing for fully flexible heterogeneity in markdowns.

I estimate the production function using the former approach with the benefit of allowing for endogenous exit (Olley and Pakes, 1996; Levinsohn and Petrin, 2003; Ackerberg, Caves and Frazer, 2015). I use cane as the flexible input for first stage inversion, and include cane price as an argument in the first stage regression, as explained below. I include the dynamic panel approach following Blundell and Bond (2000) as a robustness check in the Appendix.

First stage. The cane demand derived in (11) depends on cane price, sugar price, sugar recovery, and all other input prices and quantities, which are all observed. Cane demand

also depends on the output elasticities of labour and capital, the markdown and on productivity, which are all latent.

$$M_{it} = m\left(P_t^s, W_{it}^M, W_{it}^L, K_{it}, \beta^L, \beta^K, \beta_{it}^M, \Omega_{it}, \mu_{it}\right) \tag{21}$$

The output elasticities β^L and β^K are assumed to be the same for all mills and time periods. Markdowns and productivity vary, however, across mills and time. Without making further assumptions on the distribution of at least one of these two variables, the scalar unobservable assumption is violated, and Ackerberg, Caves and Frazer (2015) is not identified. Input demand can vary due to productivity and/or markdowns.

However, given the nested supply model for cane, all variation in markdowns is captured by observed sugar and cane prices, and cane market shares. A similar assumption was made in De Loecker et al. (2016) for a setting in which there is only imperfect competition downstream. Therefore, I include sugar price P_{it}^s, cane price W_{it}^M, and market shares $s_{it} \equiv (s_{j|0}, s_0)$ in the first stage regression, which is given by (22). The market shares included are total cane market share, and the inner-nest cane market share. Measurement error ε_{it}^q is assumed to be additive to the function $\phi_t(\cdot)$.

$$q_{it} = \phi_t\left(\tilde{l}_{it}, k_{it}, W_{it}^M, W_{it}^L, P_{it}^s, \beta_{it}^M, s_{it}\right) + \varepsilon_{it}^q \tag{22}$$

I estimate this model using a second-order polynomial in all inputs to purge measurement error and recover fitted $\hat{\phi}_{it}$. Productivity can then be recovered as a function of the data and estimable parameters.

$$\omega_{it} = \hat{\phi}_{it} - h(\tilde{l}_{it}, k_{it}, \beta) - a(W_{it}^L) \tag{23}$$

Second stage. I allow for endogenous exit by estimating an exit probability $\hat{p}_{it}^x$ using a probit regression of observed exit on all the variables in the input demand function. I allow for this exit probability to be endogenous to productivity, as proposed by Olley and Pakes (1996), and following the implementation of Collard-Wexler and De Loecker (2015). I also allow productivity to depend on a drought dummy, $\mathbf{D}_{it}$, to reflect changes in local market structure as in De Loecker (2013) and Braguinsky et al. (2015).

$$\omega_{it} = g\left(\omega_{i,t-1}, \hat{p}_{i,t-1}^x, \mathbf{D}_{it}\right) + \nu_{it} \tag{24}$$

The productivity innovation ν_{it} is given by the difference between productivity and its

expected value from the law of motion.

$$\nu_{it} = \omega_{it} - \mathbb{E}(\omega_{it}|\omega_{it-1}) \tag{25}$$

Since labour is static and capital is predetermined, the innovation ν_{it} is orthogonal to lagged labour and current capital, giving the moment conditions

$$\mathbb{E}\left[\nu_{it}\left(\tilde{l}_{i,t-1},\, k_{it},\, w^{L}_{it}\right)\right] = 0 \tag{26}$$

I impose a third-order polynomial function for ω_{it-1} and a linear function for $\hat{p}^{x}_{i,t-1}$ and Z_{it} in the productivity law of motion, and block-bootstrap standard errors with 200 iterations.

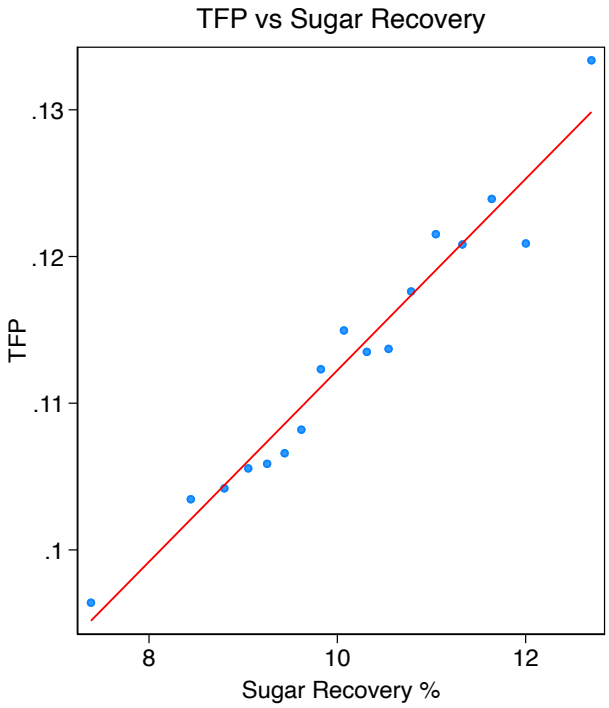
Results. The estimated output elasticities are in Table 3 below. The estimates using the control function approach are in Panel B, and are 0.715 and 0.317 for labour and capital respectively. These estimates are respectively lower and higher compared to the OLS estimates in Panel A, which are 0.740 and 0.285 for labour and capital. Both specifications have a scale parameter that is just above one. Figure 20 plots estimated TFP against recovery, highlighting a strong positive relationship between the two productivity measures.

Table 3: Production Function Estimates

	A. Ordinary Least Squares		**B. Control Function**	
	Estimate	SE	Estimate	SE
Output Elasticity of Labour	0.740	0.012	0.715	0.064
Output Elasticity of Capital	0.285	0.004	0.317	0.022
Scale Parameter	1.025		1.032	
Observations	10,620		5,788	

Notes: Standard errors are block-bootstrapped with 200 iterations.

Figure 20: TFP and sugar recovery exhibit a strong positive relationship



Notes: Figure shows a binscatter plot for estimated TFP using the control function approach following Ackerberg, Caves and Frazer (2015) against sugar recovery.

7.2 Supply Elasticities and Markdowns

Results. Panel A of Table 4 shows OLS and IV estimates for α/σ, the substitution elasticity between mills. Panel B shows estimates for θ, the substitution elasticity between crops. As expected, mills are closer substitutes for one another than cane is for competing crops, suggesting substantial crop switching costs. Note that the cross-crop elasticity estimate is in line with values from the agricultural trade literature, which typically finds substitution elasticities across commodities ranging between 1.5 and 3 (Bergquist et al., 2022; Costinot, Donaldson and Smith, 2016; Dominguez-Iino, 2026; Sotelo, 2020). Together, the two elasticities pin down the dissimilarity $\rho_0 = \frac{\theta\sigma}{\alpha}$ at 0.34, which lies between 0 and 1, thereby satisfying the consistency of the two nests.

Table 4: Supply-side Substitution Elasticities

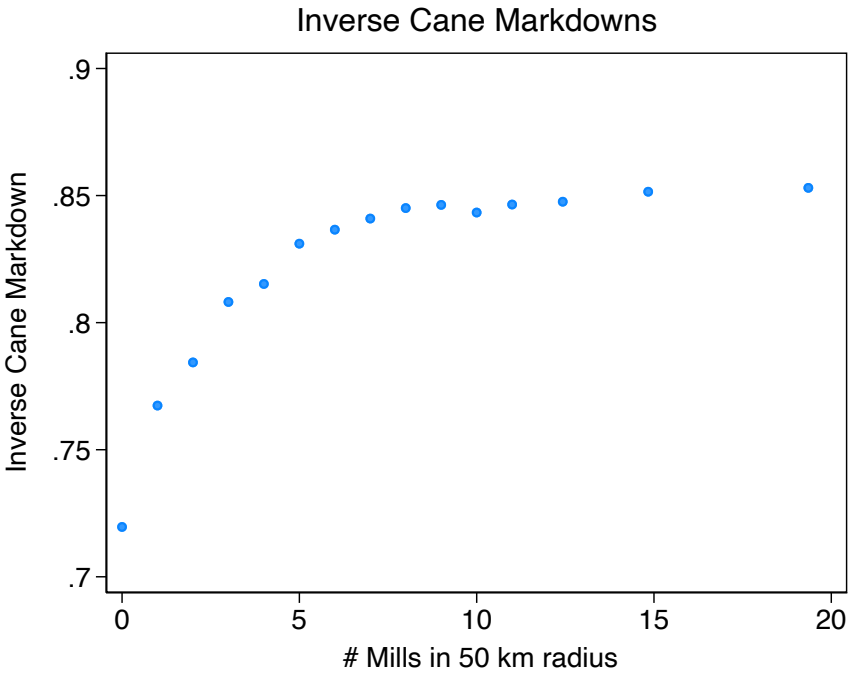
	A. Cross Mill Substitution		**B. Cross Crop Substitution**	
	OLS	IV	OLS	IV
α/σ	1.067***	6.204***		
	(0.176)	(1.217)		
θ			0.450***	2.120***
			(0.061)	(0.340)
District FE	✓	✓	✓	✓
State-Year FE	✓	✓	✓	✓
Observations	8,618	8,618	21,178	21,178
KP Wald F-statistic		152.1		42.5

Markdowns. Estimated markdown is 1.25 at the average sugar mill, and 1.19 at the median sugar mill. This implies that farmers selling to the average mill receive around 80% of cane's marginal revenue product. This number is largely comparable to figures for other agricultural goods in the literature as can be seen in Table 5. Figure 21 shows how inverse cane markdowns vary as a function of local market structure. Farmers in near-monopsony receive 72% of cane's marginal revenue product, while this figure exceeds 85% in markets with more than 15 mills.

Table 5: Average Markdowns in Agricultural Goods

Paper	Markdown	Input
Rubens (2023)	2.90	Tobacco Leaf
Zavala (2022)	1.64	Various Crops
Dominguez-Iino (2026)	1.35	Beef
This Paper	**1.25**	**Sugarcane**
Avignon and Guigue (2025)	1.19	Raw Milk
Dominguez-Iino (2026)	1.16	Maize, Soybean
Crespi and Sexton (2005)	1.10	Cattle, Potato, Rice

Figure 21: Farmers in competitive markets receive a larger share of cane's MRP



Notes: Figure shows a binscatter plot for inverse cane markdowns against local market structure – defined by the number of competing mills located within a mill's catchment of 50 kilometres.

8 Counterfactuals (Under Construction)

The estimated model enables three sets of counterfactual exercises: a welfare decomposition that separates the between-firm and within-firm channels, and two policy counterfactuals that evaluate interventions to mitigate the distributional cost of restructuring.

8.1 Welfare Decomposition

The central quantitative exercise separates the aggregate productivity gain from drought-induced restructuring into the two channels the model distinguishes: a between-firm component, from the reallocation of cane toward surviving mills, and a within-firm component, from the upgrading those survivors undertake. I define four scenarios. In the observed post-drought outcome (CF1), both exit and upgrading have occurred. In the no-drought benchmark (CF2), market structure and recovery remain at their pre-drought levels. In the intermediate scenario (CF3), mills exit and cane is reallocated to survivors, but recovery is held at its pre-drought level, shutting down the upgrading response. Comparing CF1 to CF2 gives the total effect of restructuring; CF3 relative to CF2 iso-

lates the between-firm reallocation channel; and CF1 relative to CF3 isolates the within-firm upgrading that reallocation triggers. Because the effort first-order condition makes upgrading increasing in processed volume, the gap between CF1 and CF3 measures precisely the adaptation that the reallocation windfall induces. For each welfare object $W \in \{\text{Total}, \text{Mill}, \text{Farmer}\}$:

$$\underbrace{W^{\text{CF1}} - W^{\text{CF2}}}_{\text{Total effect}} = \underbrace{W^{\text{CF3}} - W^{\text{CF2}}}_{\text{(A) Between-firm}} + \underbrace{\text{Residual} = W^{\text{CF1}} - W^{\text{CF3}}}_{\text{(B) Within-firm}} \tag{27}$$

I pair the decomposition with a full welfare accounting for both sides of the market, tracing how each channel maps into mill profits and farmer surplus. On the mill side, I compute variable profits net of upgrading costs across scenarios. On the farmer side, I evaluate surplus through the inclusive value of the nested supply system, which aggregates the prices, distances, and amenities each farmer faces over the available mills. This accounting makes visible the tension at the heart of the paper. The between-firm and within-firm channels both raise aggregate productivity, yet the same restructuring that generates them lowers the residual cane-supply elasticity each mill faces and widens its markdown. The productivity gains therefore accrue largely to mills, while farmer surplus, governed by the cane prices that do not rise, does not see benefits. Aggregate efficiency and its distribution are not separate outcomes but joint products of the same force.

8.2 Policy Counterfactuals

Next, I use the estimated model to evaluate interventions aimed at protecting farmers without foregoing the adaptation that restructuring produces.

Price floors. Avignon and Guigue (2025) note that a price floor on raw materials can be efficient in sectors where farmers face buyers with monopsony power. The first counterfactual I study targets such a price floor and its recovery premium. The floor raises the price a mill must pay as its recovery rises, so it protects farmer prices directly; but for the same reason it blunts the incentive to upgrade, since a higher recovery raises the mill's own cane bill through the premium – the suppression channel that appears as the second term in the effort first-order condition. I trace out farmer surplus, mill profits, and aggregate recovery as I vary the base price and the recovery premium, mapping the frontier along which price support and upgrading trade off. The exercise quantifies how much adaptation is sacrificed for a given gain in farmer prices, and whether a premium can be

set that supports farmers while leaving the upgrading incentive largely intact.

Upgrading cost subsidies. The second intervention subsidizes upgrading directly, lowering the marginal cost of raising recovery through a per-unit or ad valorem subsidy on effort. Where the price floor works against upgrading, the subsidy works with it, delivering the recovery gains without relying on the concentration that funds them in the drought-driven equilibrium. The key question is distributional: because a subsidy leaves the residual-supply elasticity and hence the markdown untouched, whether its benefits reach farmers, rather than being retained by mills exactly as the concentration-driven gains were, is not obvious a priori. I evaluate the subsidy's effect on recovery, mill profits, and farmer surplus, and – holding government expenditure fixed – compare it against the price floor as an alternative route to the same recovery gain. The comparison identifies which instrument can loosen the link between adaptation and its distribution, the policy question the paper's central tension raises.

9 Conclusion

This paper shows that climate change reshapes agricultural supply chains not only by lowering what farmers grow, but by restructuring the markets in which they sell. In the Indian sugar industry, extreme droughts drive the least productive mills to exit and reallocate their cane to nearby survivors. This reallocation is not merely compositional: by expanding the volume base over which quality improvements apply, it raises survivors' return to upgrading and triggers within-firm productivity growth. But the same concentration that funds adaptation also widens mills' buyer power: farm-gate cane prices do not rise, so the productivity gains are captured by mills rather than shared with the millions of farmers who supply the industry. Climate-induced adaptation and exploitation are thus complements, driven by a single change in local market structure.

These findings carry a broader lesson for how we think about climate change and agricultural markets. The welfare consequences of a climate shock for smallholders depend not only on its direct effect on yields, but on how it reshapes the market power of the intermediaries they depend on — a channel largely absent from existing work, which focuses on direct weather-to-yield effects. Where processors adapt, aggregate efficiency can rise even as farmers are left worse off, so the efficiency gains from climate-induced restructuring and its distributional costs are two faces of the same mechanism. This poses a genuine policy trade-off: interventions that prevent exit forgo both the reallocation gains and the

within-firm upgrading that reallocation enables, while doing nothing leaves farmers exposed to widening markdowns.

My counterfactual analysis suggests that policies targeting the distributional margin directly — sustaining competition for cane or linking farm-gate prices to recovery — can protect farmers while preserving the adaptation that concentration sets in motion. As climate shocks become more frequent, understanding how they redraw the boundaries of market power, and not only the productivity of the land, becomes central to protecting smallholders who remain at the base of global agricultural supply chains.

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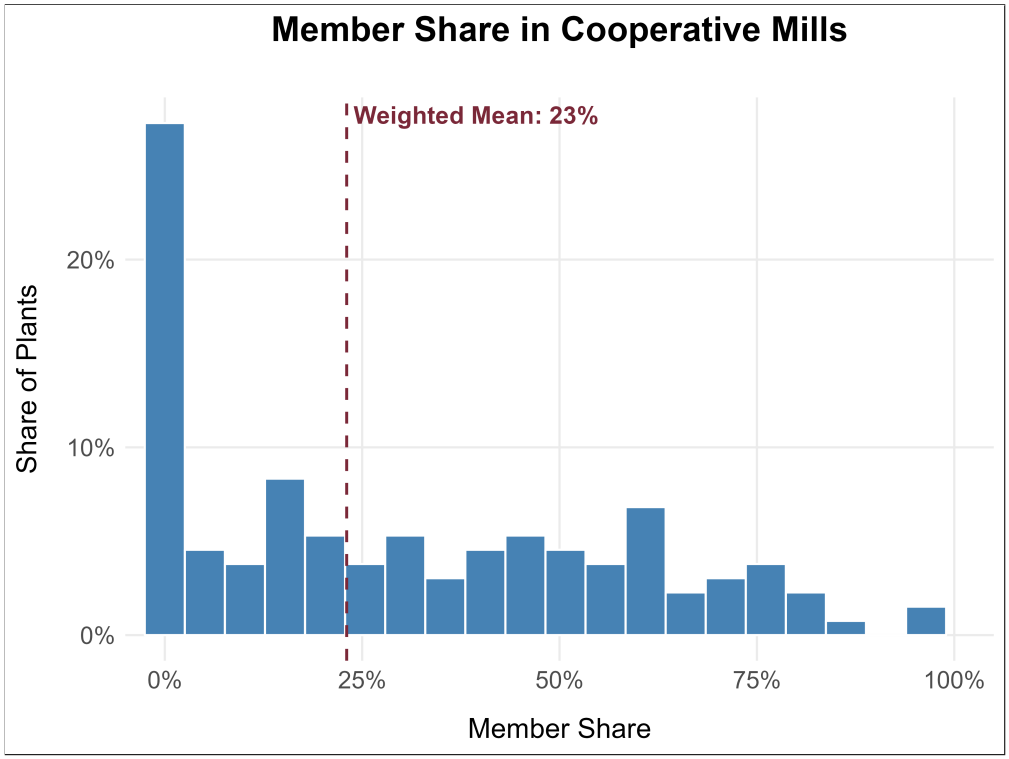
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A Appendix

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A.1 Figures

Figure A.1: Share of Members in Cooperative Mills



Notes: This figure plots the share of member farmers in cooperative mills across Maharashtra in 2025-26. *Source:* Farmer-Mill Transactions and Plant-level Records from the Maharashtra Sugar Commissionerate.

Figure A.2: Industrial Water Requirement

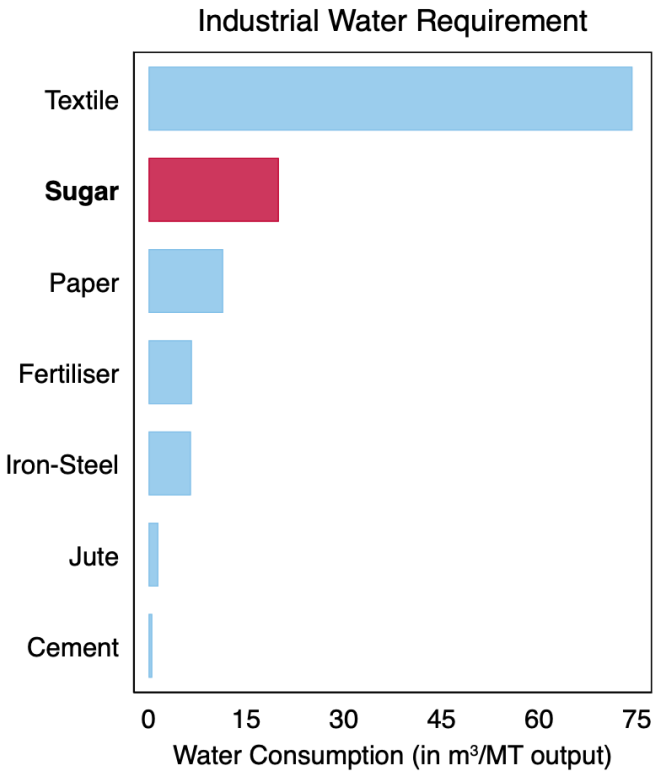
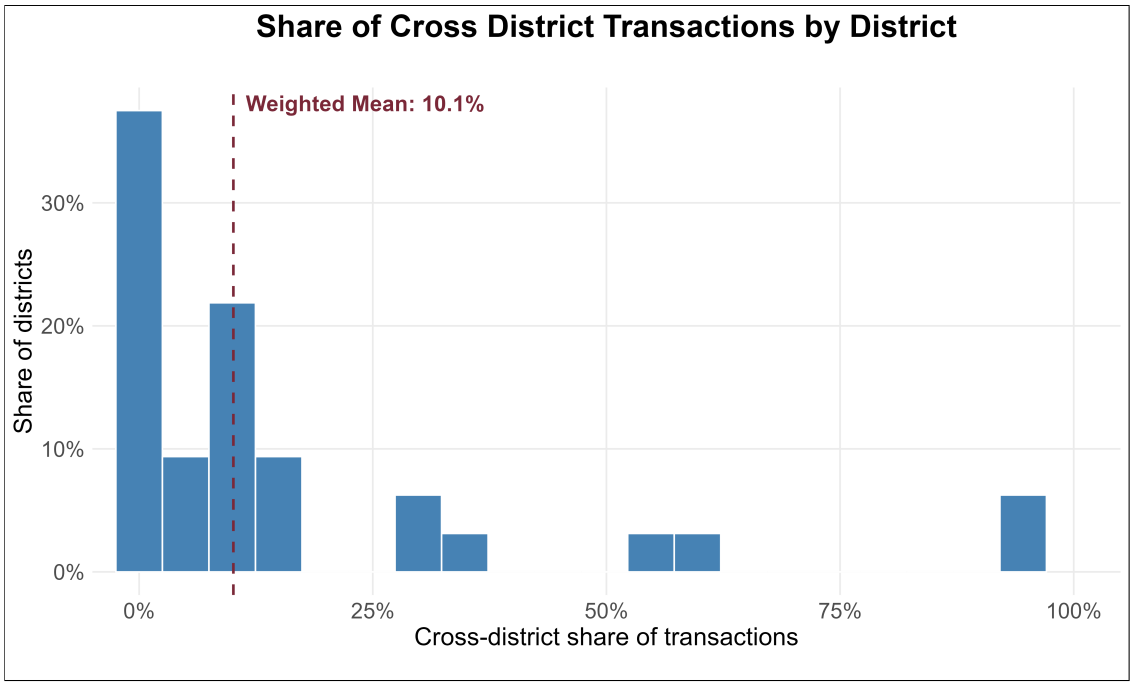


Figure A.3: Newspaper Headlines on Drought-induced Mill Closures



Figure A.4: Share of Cross District Farmer-Mill Transactions



Notes: This figure plots the share of farmer-mill transactions that take place across district boundaries. *Source:* Farmer-Mill Transactions and Plant-level Records from the Maharashtra Sugar Commissionerate.

Figure A.5: Monthly Sugar Release Allocations

Sl. No.	Plant Code	Plant Name	Notional Stock as on 31.07.2024	Quota calculated as per Notional Stock
(1)	(2)	(3)	(4)	(5)
	State:	**Punjab**		
383	101	GURDASPUR	11466	2371
384	201	BATALA	9126	1887
385	301	FAZILKA	3111	643
386	401	MORINDA	15514	3208
387	501	BHOGPUR	17270	3571

Notes: This figure shows monthly sugar release allocations outlined by the Indian Directorate of Sugar for sugar mills in Punjab for August 2024. *Source:* Indian Directorate of Sugar.

A.2 Tables

A.3 Theoretical Appendix